Appendix C

Atlas of comparison spectra

C.1 Introduction

This Appendix presents the atlas of comparison spectra useful for wavelength calibration of the AFOSC spectra and the tables with the recommended exposure times. The atlas is made using the mechanical mounting of the previous detector. The lines of comparision spectra are shifted ~ 50 pixels for the actual mounting. Exposure times are calculated to obtain 40000–50000 counts for the brightest line in the spectrum (except when indicated). They have to be considered indicative, since they are subjected to intrinsic intensity variations, lamp substutitions, different positionig of the lamps in their holders, etc.

As described in Sect. 3.3.4, a single lamp spectrum usually does not allow a good calibration due to the incomplete wavelength coverage and/or insufficient number of lines. For this reason we present the atlases of single lamp as well as coadded spectra of two lamps. All the spectra shown in this Appendix were taken with the 0.7 arcsec slit. This made easier the evaluation of possible blends when calibrating spectra taken with a wider slit.

The spectral range might show some changes, since shifts up to 50 pixels could be present between spectra taken with different slits. Run to run difference of spectra taken with the same slit are instead much smaller, typically within 2-3 pixels, if a given slit was permanently mounted on the instrument.

Tables C.1–C.3 report the list of lines of He, Ne, Ar and Hg-Cd lamps. They were taken from the FORS User Manual (left column) and IRAF linelist (right column). Some minor discrepancies are present between the two linelists. The Th lamp atlas was prepared by comparing the AFOSC spectra with the atlas of high resolution spectrographs like SARG and UVES. Only the lines that are relatively unblended are included in the AFOSC atlas.

The Neon lamp used for the preparation of the atlas presented here failed in July 2001. Exposure times and spectral lines intensities of the new lamp are similar to the older one.

Wavelength (Å)	Element	Wavelength (Å)	Element
		3187.743	He I
		3464.14	Ar II
		3520.472	Ne I blend with Ar 3520.0
		3545.58	Ar II
		3559.51	Ar II
3610.5000	Cd		
3650.1440	Hg		
3663.2739	Hg		
		3718.21	Ar II blend with Ar II 3724.51
		3780.84	Ar II
		3850.57	Ar II
3888.6460	He I	3888.646	He I
		3928.62	Ar II
		3964.727	He I
		4026.189	He I
		4044.418	Ar I
4046.5569	Hg		
		4072.2	Ar I blend 4072.01 4072.40
4077.8313	Hg		
		4131.73	Ar II
		4158.590	Ar II
		4259.361	Ar II
		4277.55	Ar II blend with Ar I 4272.168
		4300.4	Ar II blend
4358.3428	Hg		
		4426.01	Ar II
4471.4790	He I	4471.477	He I
		4510.733	Ar I
		4545.08	Ar II
		4579.39	Ar II
		4657.94	Ar II
4678.1602	Cd		
		4713.143	He I
		4764.89	Ar II
4799.9199	Cd		
		4806.07	Ar II
		4879.90	Ar II
4921.9292	He I	4921.929	He I
		4965.12	Ar II
5015.6748	He I	5015.675	He I
5085.8242	Cd		
		5187.746	Ar I
		5221.270	Ar I
5400.5620	Ne I	5400.562	Ne I
5460.7422	Hg		
		5495.872	Ar I
		5572.548	Ar I
		5606.732	Ar I
		5650.703	Ar I
		5748.299	Ne I
5764.4189	Ne I	5764.418	Ne I
5769.5981	Hg		
5790.6558	Hg		
		5852.4878	Ne I
5875.6201	He I	5875.618	He I
5944.8301	Ne I	5944.8342	Ne I
5975.5342	Ne I		

Table C.1: Wavelength of the lines of He, Ne, Ar and Hg-Cd lines.

Wavelength (Å)	Element		Wavelength (Å)	Element
6029.9971	Ne I		6029.9971	Ne I
6074.3379	Ne I		6074.3377	Ne I
6096.1602	Ne I		6096.1630	Ne I
6143.0625	Ne I		6143.0623	Ne I
6163.5938	Ne I		6163.5939	Ne I
6217.2812	Ne I		6217.2813	Ne I
6266.4951	Ne I		6266.4950	Ne I
6304.7900	Ne I		6304.7892	Ne I
6334.4277	Ne I		6334.4279	Ne I
6382.9912	Ne I		6382.9914	Ne I
6402.2456	Ne I		6402.246	Ne I
6438.4697	Cd			
6506.5278	Ne I		6506.5279	Ne I
6532.8799	Ne I		6532.8824	Ne I
6598.9531	Ne I		6598.9529	Ne I
6678.1489	He I		6678.149	He I
			6678.2764	Ne I
6717.0400	Ne I		6717.0428	Ne I
			6752.832	Ar I
			6871.290	Ar I
6907.1602	Hg			
6929.4678	Ne I		6929.430	Ne I
6965.4307	Ar I		6965.430	Ar I
7032.4126	Ne I		7032.4127	Ne I
			7065.188	He I
7081.8799	Hg			
7091.9902	Hg			
			7107.496	Ar I
			7125.80	Ar I
7147.0405	Ar I		7147.041	Ar I
7173.9390	Ne I		7173.939	Ne I
			7206.986	Ar I
7245.1670	Ne I		7245.167	Ne I
7272.9302	Ar I		7272.936	Ar I
7281.3491	He I		7281.349	He I
			7311.71	Ar I
7346.2002	Cd			
			7353.316	Ar I
			7372.118	Ar I
7383.8999	Cd			
7383.9805	Ar I		7383.980	Ar I
7385.2998	Cd			
7438.8999	Ne I		7438.899	Ne I
7488.8701	Ne I		7488.872	Ne I
7503.8677	Ar I		7503.867	Ar I
7514.6519	Ar I		7514.651	Ar I
			7535.775	Ne I
			7544.046	Ne I
7635.1060	Ar I		7635.105	Ar I
			7670.04	Ar I
7723.9844	Ar I		7723.8	blend 7723.760 - 7724.206
			7891.075	Ar I
7948.1763	Ar I		7948.175	Ar I

Table C.2: Wavelength of the lines of He, Ne, Ar and Hg-Cd lines.

Wavelength (Å)	Element		Wavelength (Å)	Element
8006.1567	Ar I		8006.156	Ar I
8014.7856	Ar I		8014.786	Ar I
			8053.307	Ar I
			8082.458	Ne I
8103.6929	Ar I		8103.692	Ar I
8115.3110	Ar I		8115.311	Ar I
8264.5225	Ar I		8264.521	Ar I
8300.3262	Ne I		8300.326	Ne I
8377.3672	Ne I		8377.607	Ne I
8408.2100	Ar I			
8424.6475	Ar I		8424.647	Ar I
			8495.360	Ar I
8521.4424	Ar I		8521.441	Ar I
8591.2588	Ne I			
			8605.78	Ar I
			8620.47	Ar I
8634.6484	Ne I		8634.648	Ne I
8654.3838	Ne I		8654.383	Ne I
8667.9443	Ar I		8667.943	Ar I
8704.1504	Ne I			
			8761.72	Ar I
			8780.1872	Ne I
			8783.755	Ne I
8853.8672	Ne I		8853.866	Ne I
8919.5000	Ne I			
			9075.42	Ar I
9122.9678	Ar I		9122.966	Ar I
			9194.68	Ar I
9224.4990	Ar I		9224.998	Ar I
			9291.58	Ar I
9300.8496	Ne I			
9354.2178	Ar I		9354.218	Ar I
9425.3799	Ne I		9425.38	Ne I
			9534.167	Ne I
9657.7842	Ar I		9657.874	Ar I
9784.5010	Ar I		9784.501	Ar I
10140.0000	Hg			
10394.5996	Cd			
10830.1709	He I			

Table C.3: Wavelength of the lines of He, Ne, Ar and Hg-Cd lines.

C.2 Grism #2

Lamp/Slit	0.70	0.85	1.26	1.69	2.10	3.00	4.22	8.44	Remarks
Ne	9				6				
Ar									
He	15								
Hg-Cd	90				60			20	
Th	5				3			1	only bright Ar lines

Table C.4: Exposure times (in seconds) of the calibration lamps using the Grism #2 for the available slits.

Remarks

- Zero order not shown in the atlas

Suggested lamps combinations

- Hg-Cd lamp
- He lamp

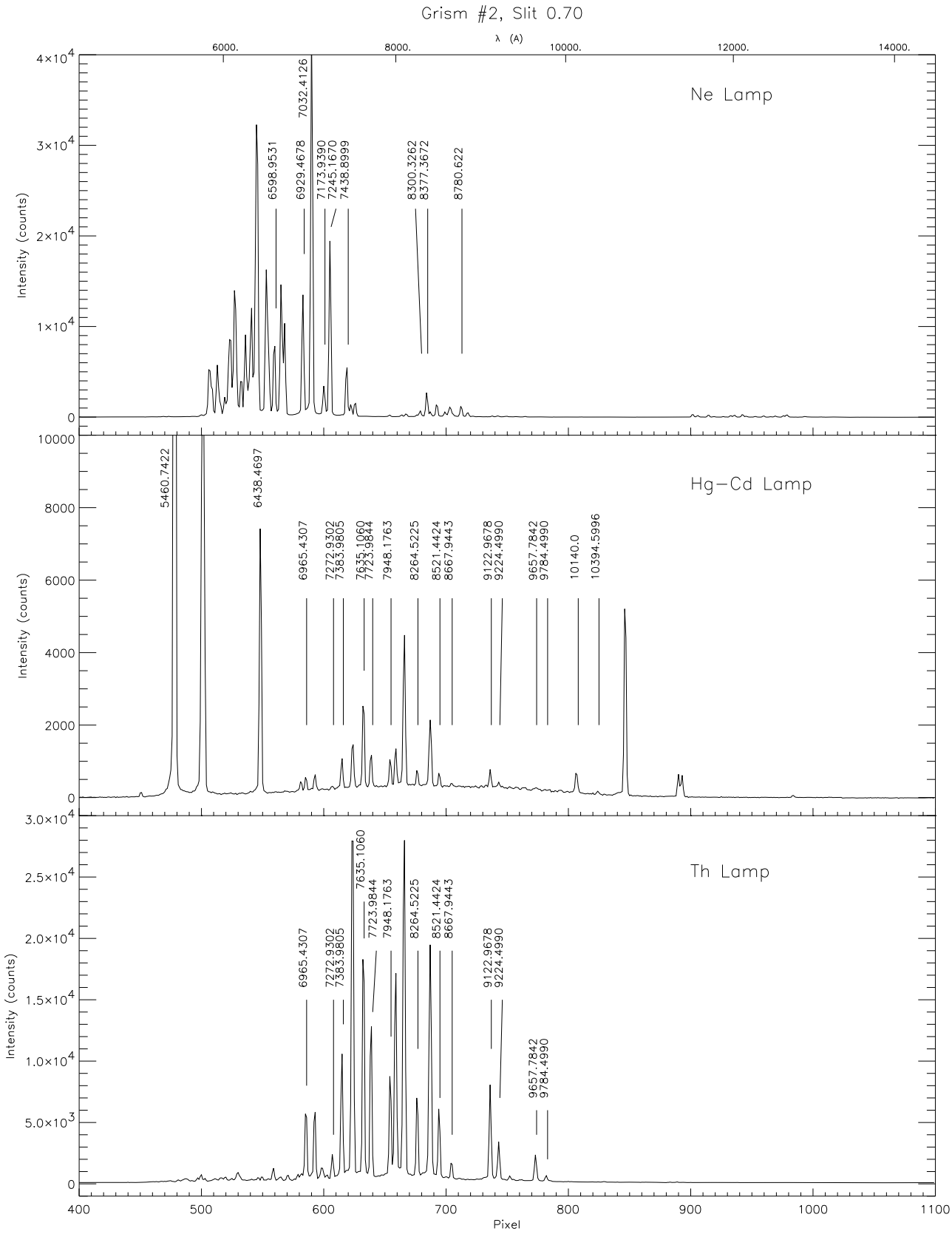


Figure C.1: Calibration spectra of Grism #2: Ne, Hg-Cd, and Th lamps

C.3 Grism #3

Lamp/Slit	0.70	0.85	1.26	1.69	2.10	3.00	4.22	8.44	Remarks
Ne	300				300				one faint line
Ar	300				300				very faint lines
He									
Hg-Cd	180				120				
Th	300				240				

Table C.5: Exposure times (in seconds) of the calibration lamps using the Grism #3 for the available slits.

Remarks

- Atlas of Ar and Ne lamps not included due to the lack of lines

Suggested lamps combinations

- Hg-Cd lamp
- He lamp
- Th lamp (when using narrow slits)

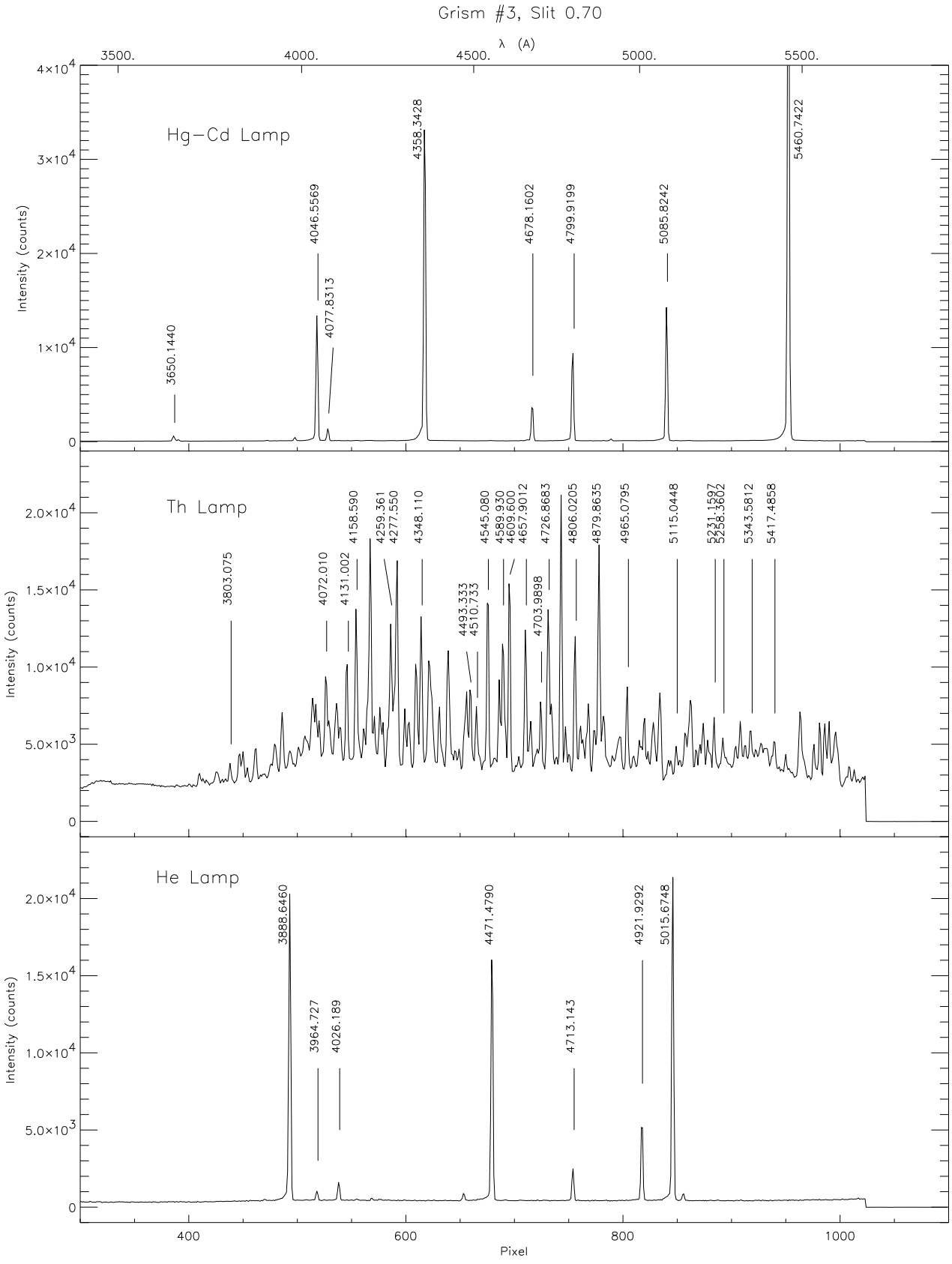


Figure C.2: Calibration spectra of Grism #3: Hg-Cd, Th and He lamps

C.4 Grism #4

Lamp/Slit	0.70	0.85	1.26	1.69	2.10	3.00	4.22	8.44	Remarks
Ne	15		12	10	10		8		
Ar	40		35	30	30				
He	20		18	16	14				
Hg-Cd	100		75	60	60	50			
Th	15		10	8	8		6		only bright Ar lines

Table C.6: Exposure times (in seconds) of the calibration lamps using the Grism #4 for the available slits.

Remarks

- Severe line blending with Th lamp

Suggested lamps combinations

- Ne + Hg-Cd lamps
- He + Ne lamps
- He + Ar lamps

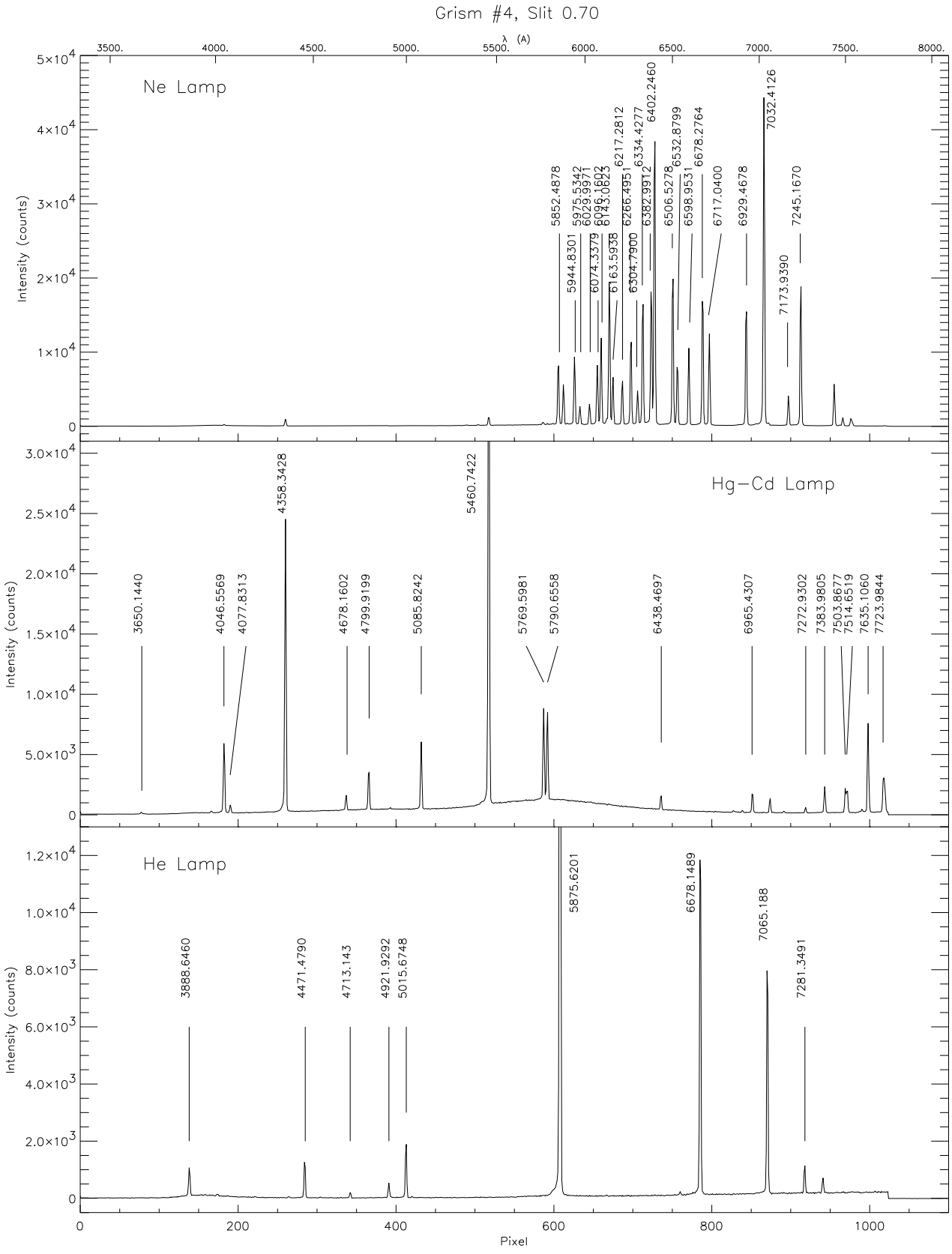


Figure C.3: Calibration spectra of Grism #4: Ne, Hg-Cd, and He lamps

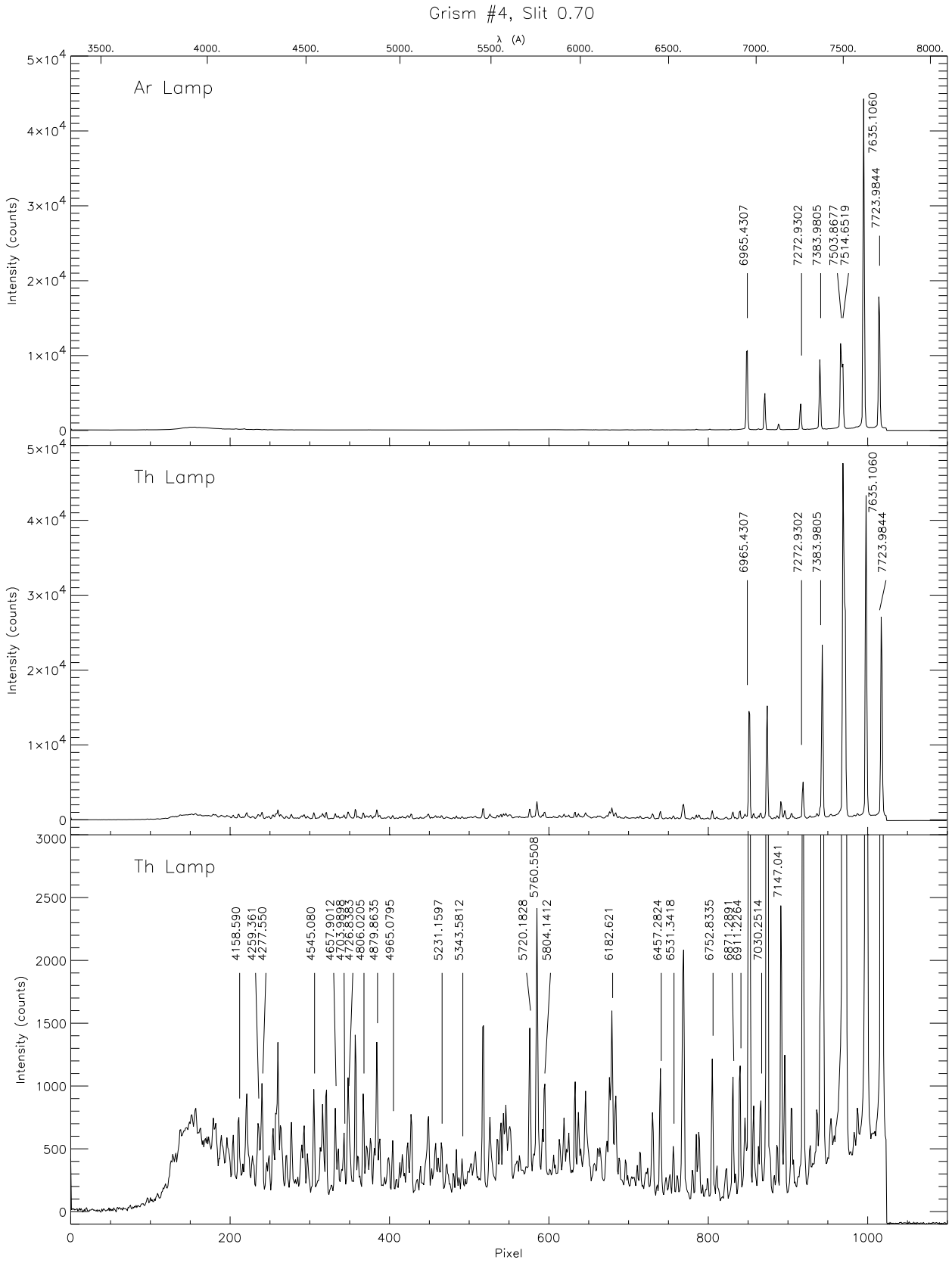


Figure C.4: Calibration spectra of Grism #4: Ar and Th lamps

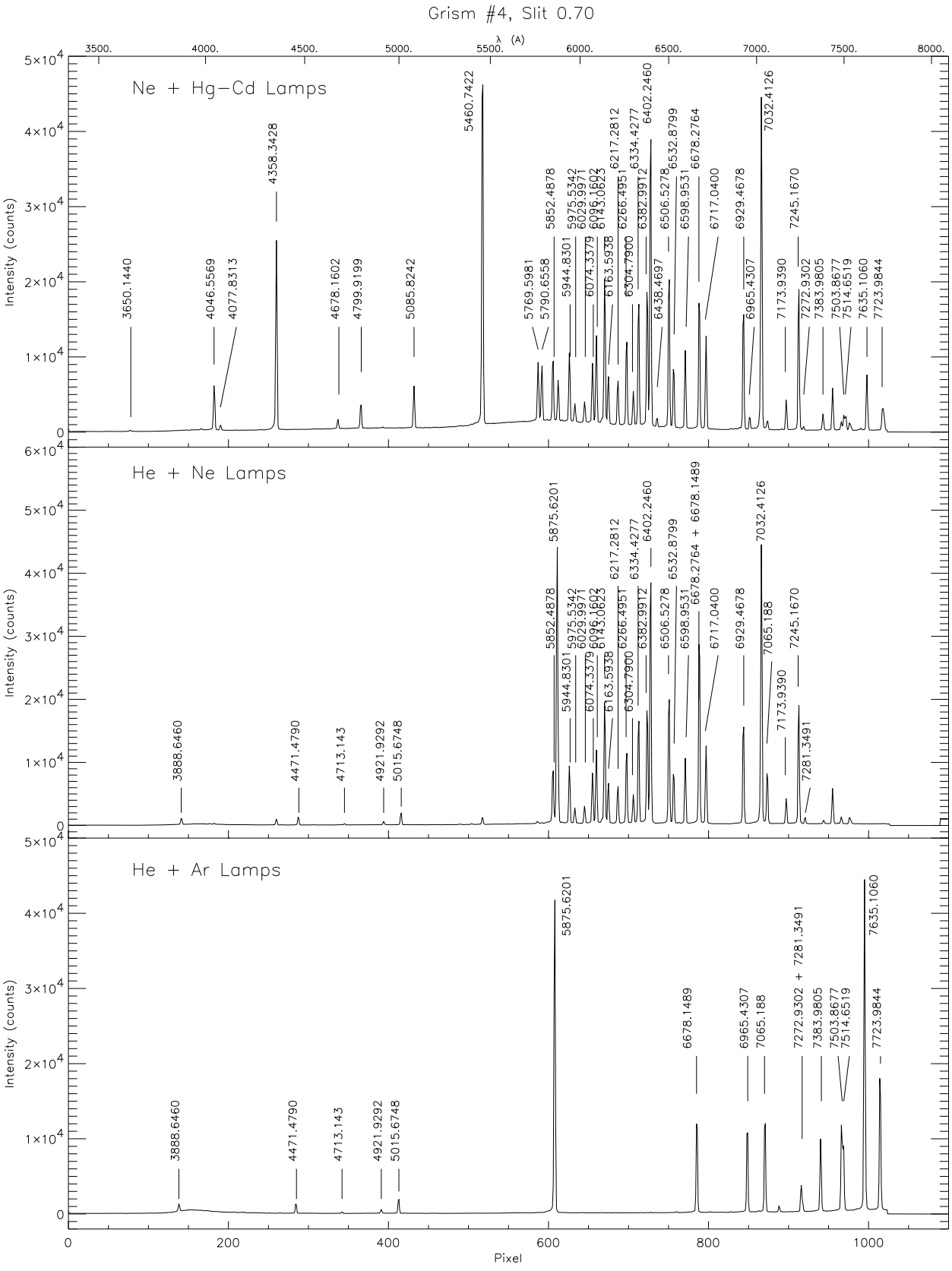


Figure C.5: Calibration spectra of Grism #4: combined spectra Ne+Hg-Cd, He+Ne and He+Ar lamps.

C.5 Grism #6

Lamp/Slit	0.70	0.85	1.26	1.69	2.10	3.00	4.22	8.44	Remarks
Ne	–	–	–	–	–	–	–	–	no lines
Ar	–	–	–	–	–	–	–	–	no lines
He	300								
Hg-Cd	150				120	120	120	80	
Th	300				240				

Table C.7: Exposure times (in seconds) of the calibration lamps using the Grism #6 for the available slits.

Remarks

- Atlas of Ar and Ne lamps not included due to the lack of lines

Suggested lamps combinations

- Hg-Cd lamp
- He lamp
- Th lamp (when using narrow slits)

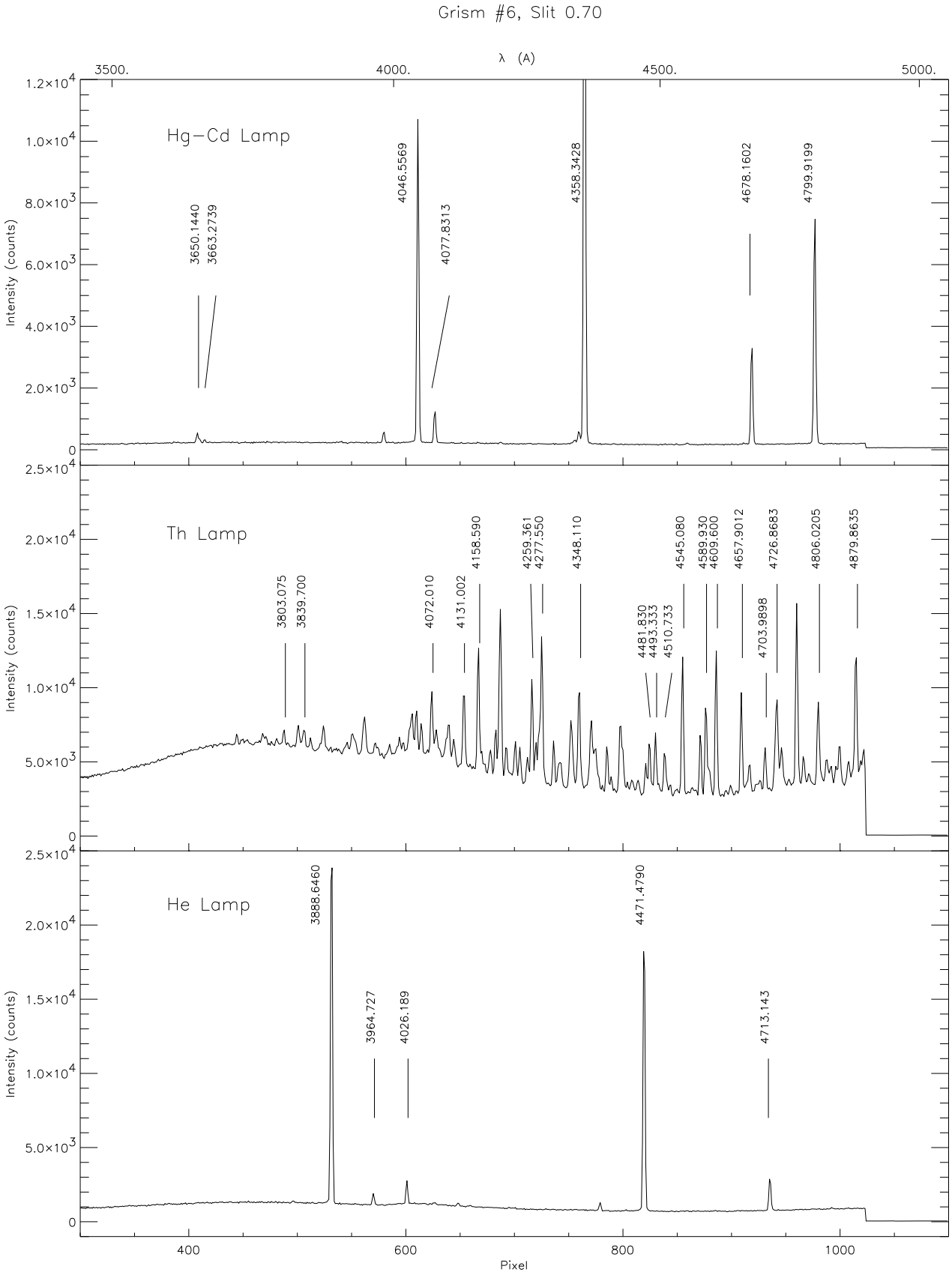


Figure C.6: Calibration spectra of Grism #6: Hg-Cd, Th and He lamps

C.6 Grism #7

Lamp/Slit	0.70	0.85	1.26	1.69	2.10	3.00	4.22	8.44	Remarks
Ne	30		25	25	20		20		
Ar	–	–	–	–	–	–	–	–	no lines
He	22		20	18	16				
Hg-Cd	150		120	120	100		90		
Th	240		300	240	240		180		

Table C.8: Exposure times (in seconds) of the calibration lamps using the Grism #7 for the available slits.

Remarks

- Atlas of Ar lamp not included due to the lack of lines

Suggested lamps combinations

- Ne + Hg-Cd lamp
- Ne + He lamp
- Th lamp (when using narrow slits)

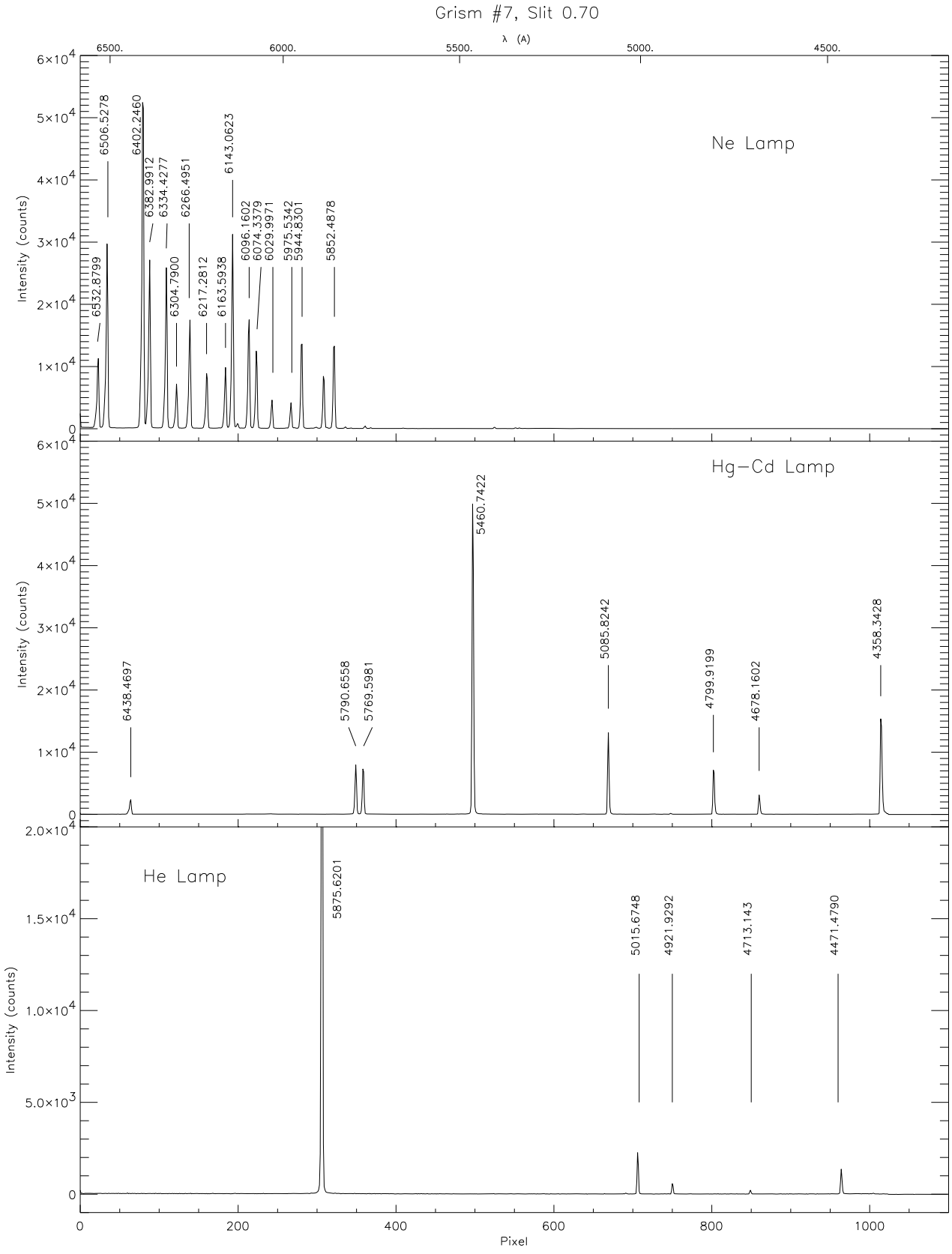


Figure C.7: Calibration spectra of Grism #7: Ne, Hg-Cd, and He lamps

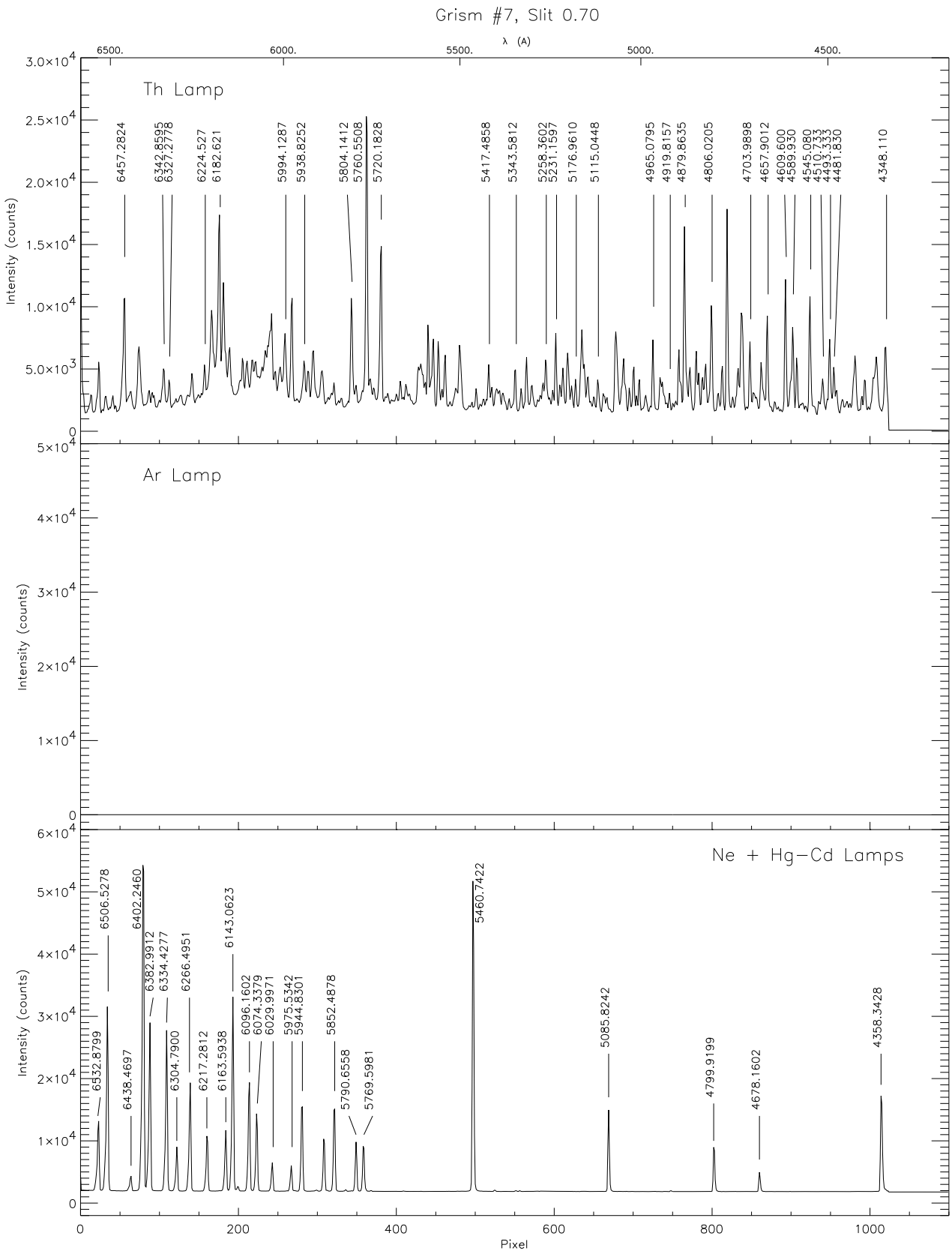


Figure C.8: Calibration spectra of Grism #7: Th, Ar and Ne+HgCd lamps

C.7 Grism #8

Lamp/Slit	0.70	0.85	1.26	1.69	2.10	3.00	4.22	8.44	Remarks
Ne	15		14	12	12		10		
Ar	35		30	25	25				
He	80	70	60	50					
Hg-Cd	300	300	300	300	300	300	300		Ar lines + 1 Hg line
Th	14		12	10	10				only bright Ar lines

Table C.9: Exposure times (in seconds) of the calibration lamps using the Grism #8 for the available slits.

Remarks

- Ar lines included in Ar, Th, and Hg-Cd lamps. Only one Cd line with the the Hg-Cd lamp

Suggested lamps combinations

- Ne + Th lamps

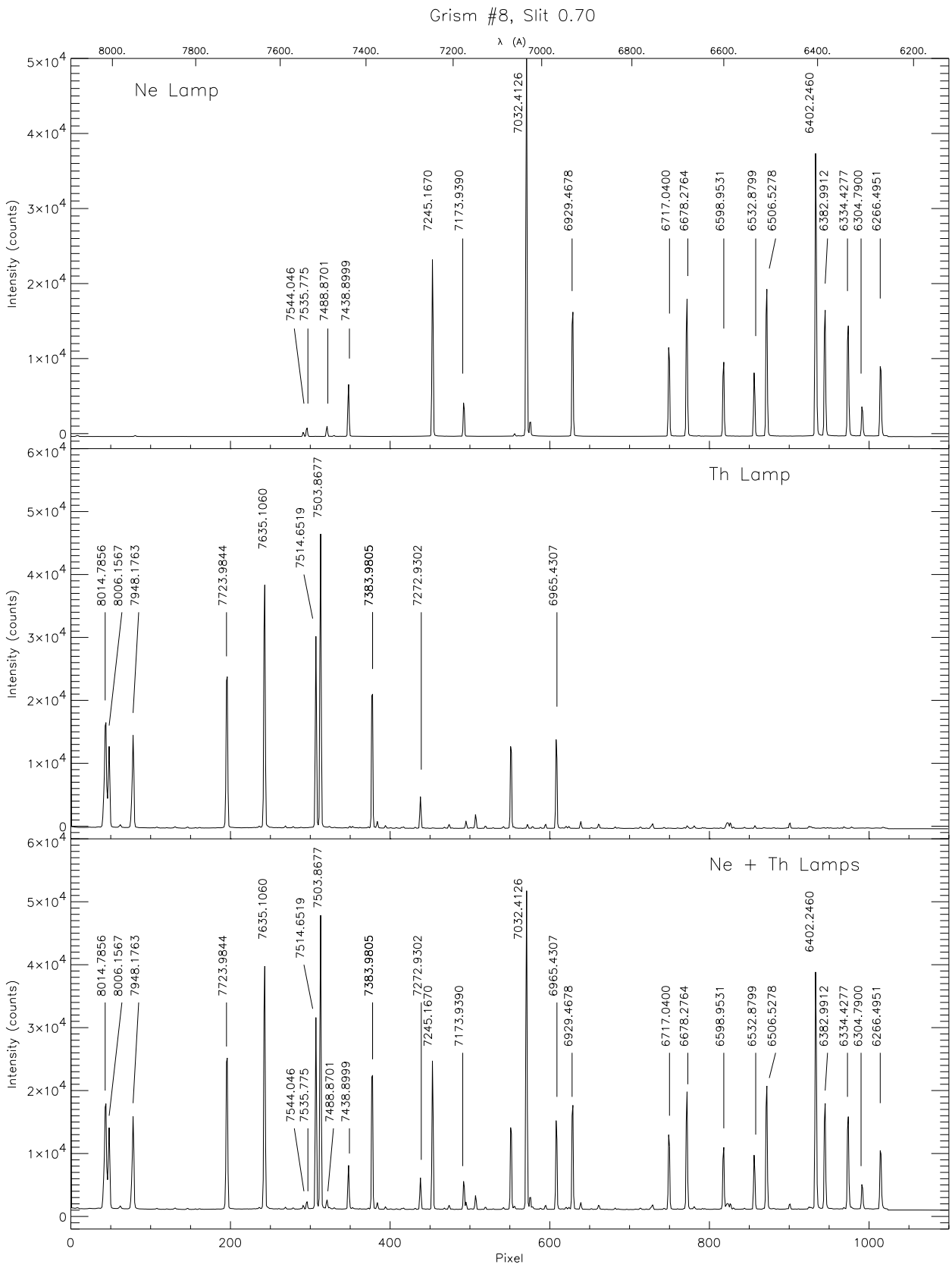


Figure C.9: Calibration spectra of Grism #8: Ne and Th lamps

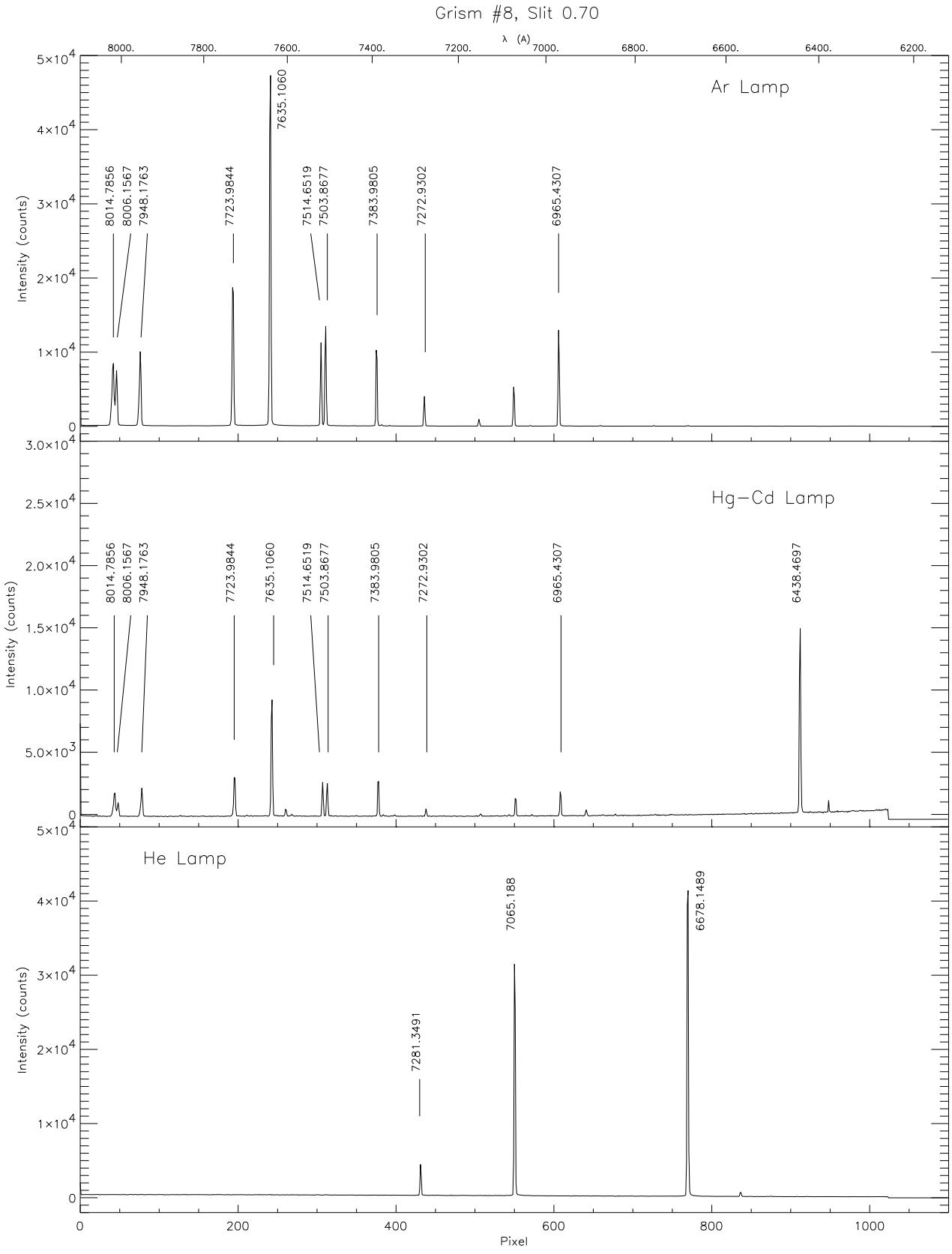


Figure C.10: Calibration spectra of Grism #8: Ar, Hg-Cd and He lamps

C.8 Grism #10

Lamp/Slit	0.70	0.85	1.26	1.69	2.10	3.00	4.22	8.44	Remarks
Ne	6		4	3	3				
Ar	8		6	5	4				
He	15		12	11	10				
Hg-Cd	70		60	50	40				
Th	2		1	1	1				only bright Ar lines

Table C.10: Exposure times (in seconds) of the calibration lamps using the Grism #10 for the available slits.

Remarks

- Zero order not shown in the atlas
- Exposure times fixed to avoid saturation of the zero order.

Suggested lamps combinations

- Hg-Cd lamp
- He lamp

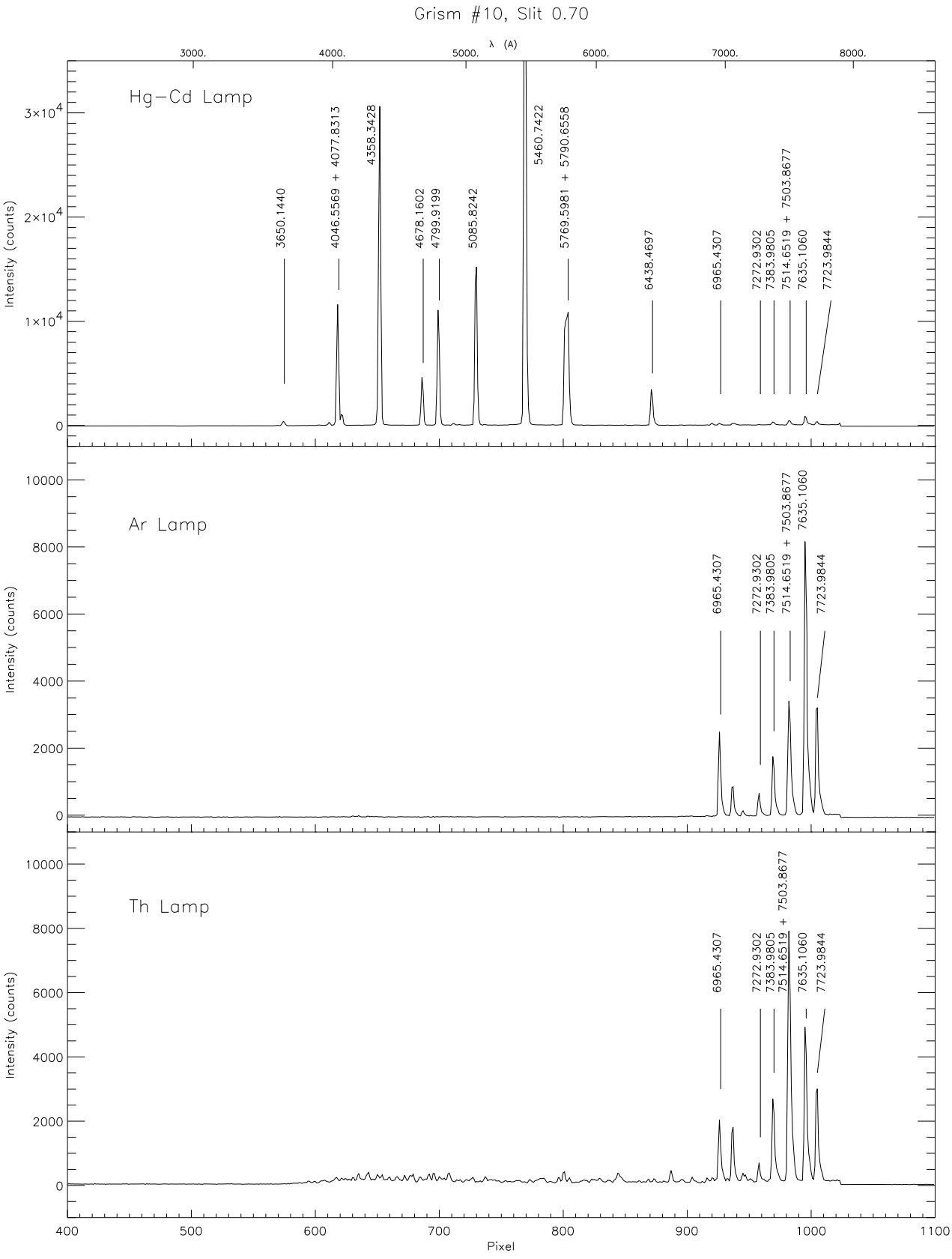


Figure C.11: Calibration spectra of Grism #10: Hg-Cd, Ar and Th lamps

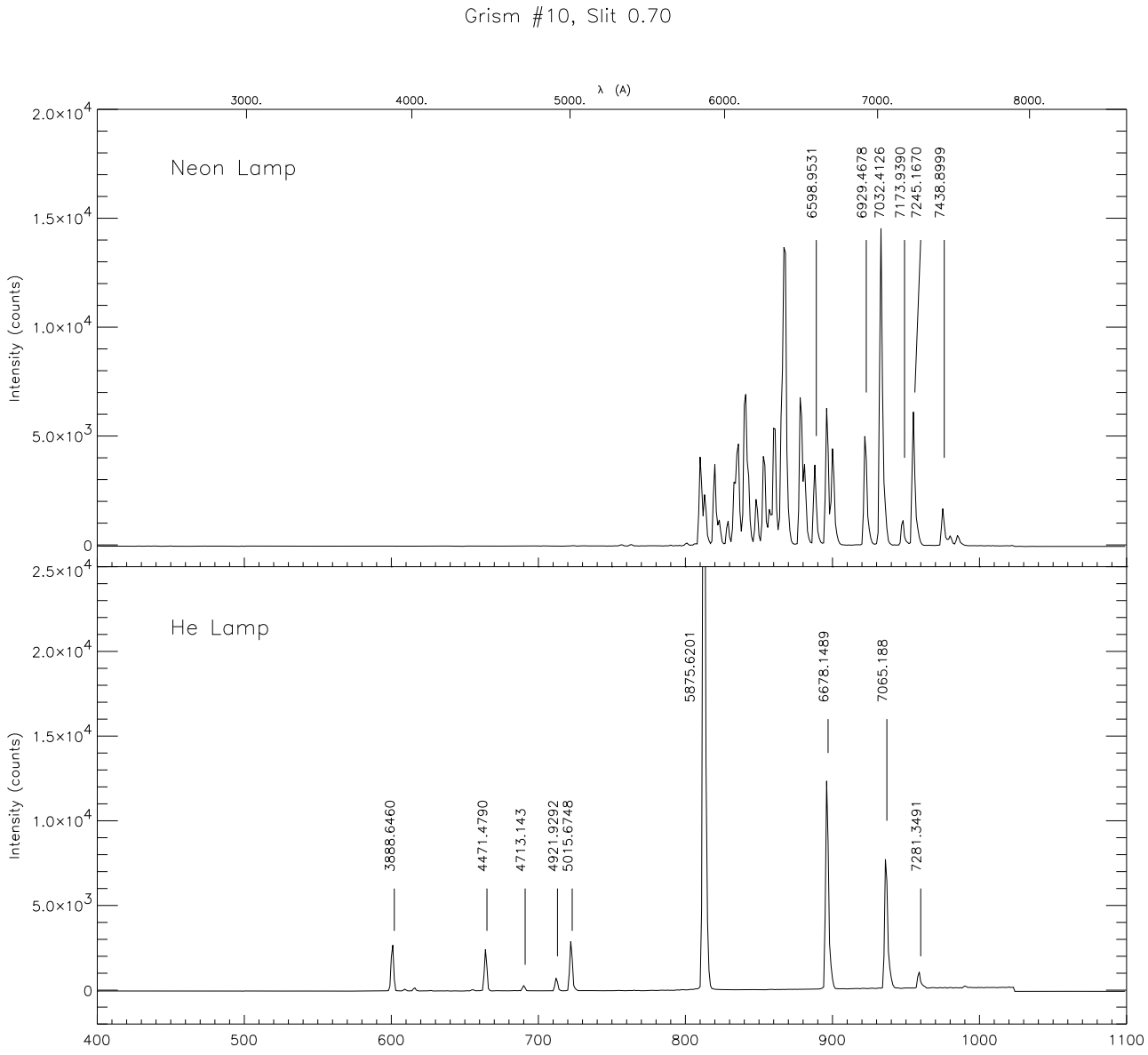


Figure C.12: Calibration spectra of Grism #10: Ne and He lamps

C.9 Grism #13

Lamp/Slit	0.70	0.85	1.26	1.69	2.10	3.00	4.22	8.44	Remarks
Ne	300	300	300	300	300	300	300	300	1 line
Ar	300	300	300	300	300	300	300	300	very faint lines
He	300								1 line
Hg-Cd	300				180				2 lines
Th	300	300	300	300	300				

Table C.11: Exposure times (in seconds) of the calibration lamps using the Grism #13 + OS1 for the available slits.

Lamp/Slit	0.70	0.85	1.26	1.69	2.10	3.00	4.22	8.44	Remarks
Ne	180				120				
Ar	60								
He	300								1 line
Hg-Cd	300	300	300	300	300	300	300	300	2 lines
Th	30				20				

Table C.12: Exposure times (in seconds) of the calibration lamps using the Grism #13 + i filter for the available slits.

Remarks

- The echelle configuration (Grism #13 + Grism #2 or #10 as cross disperser) is not considered due to its limited uselfulness

Suggested lamps combinations: Gr #13 + OS1

- Th lamp

Suggested lamps combinations: Gr #13 + i

- Ar lamp
- Th lamp

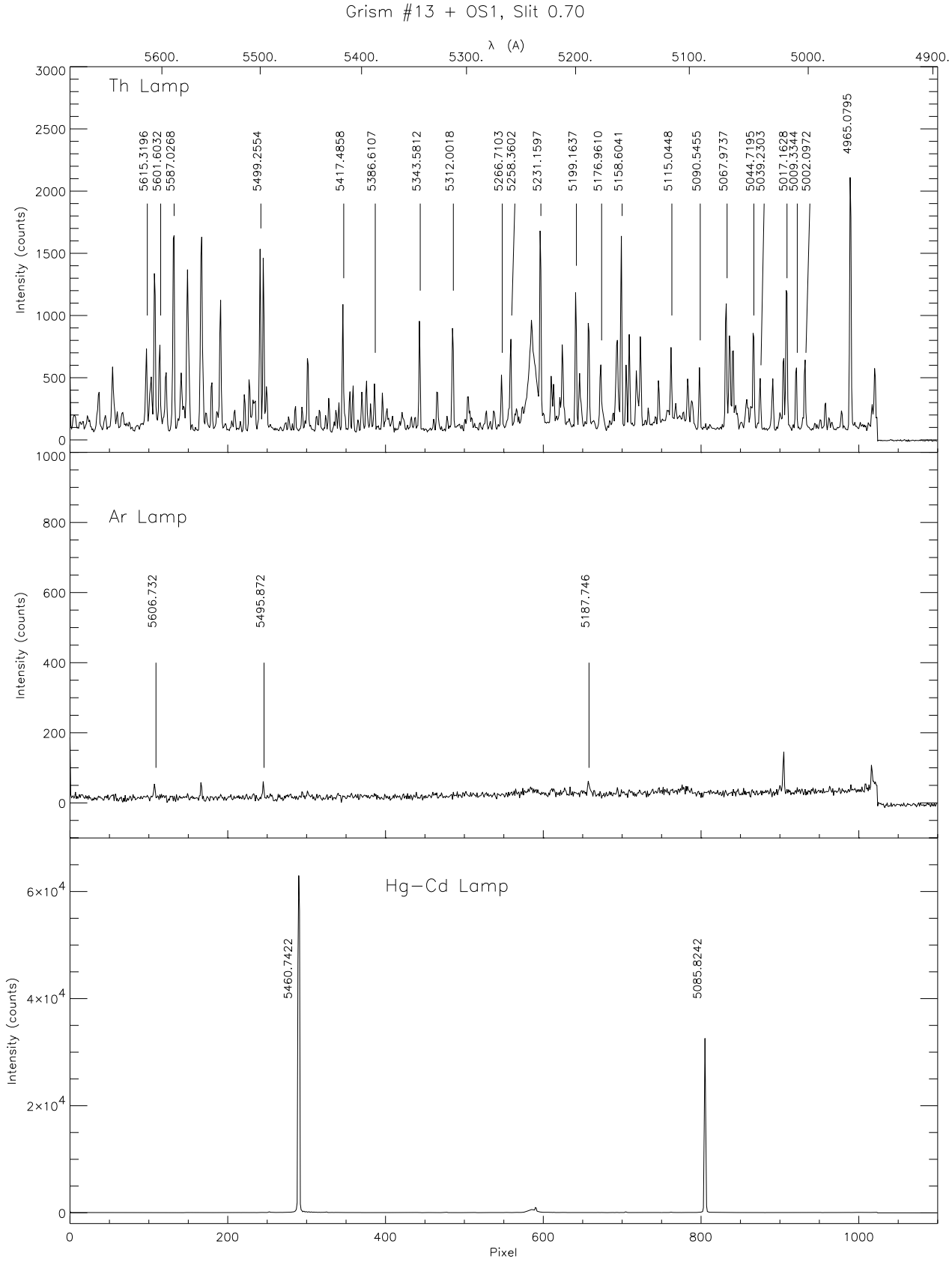


Figure C.13: Calibration spectra of Grism #13 + OS1: Th, Ar and Hg-Cd lamps

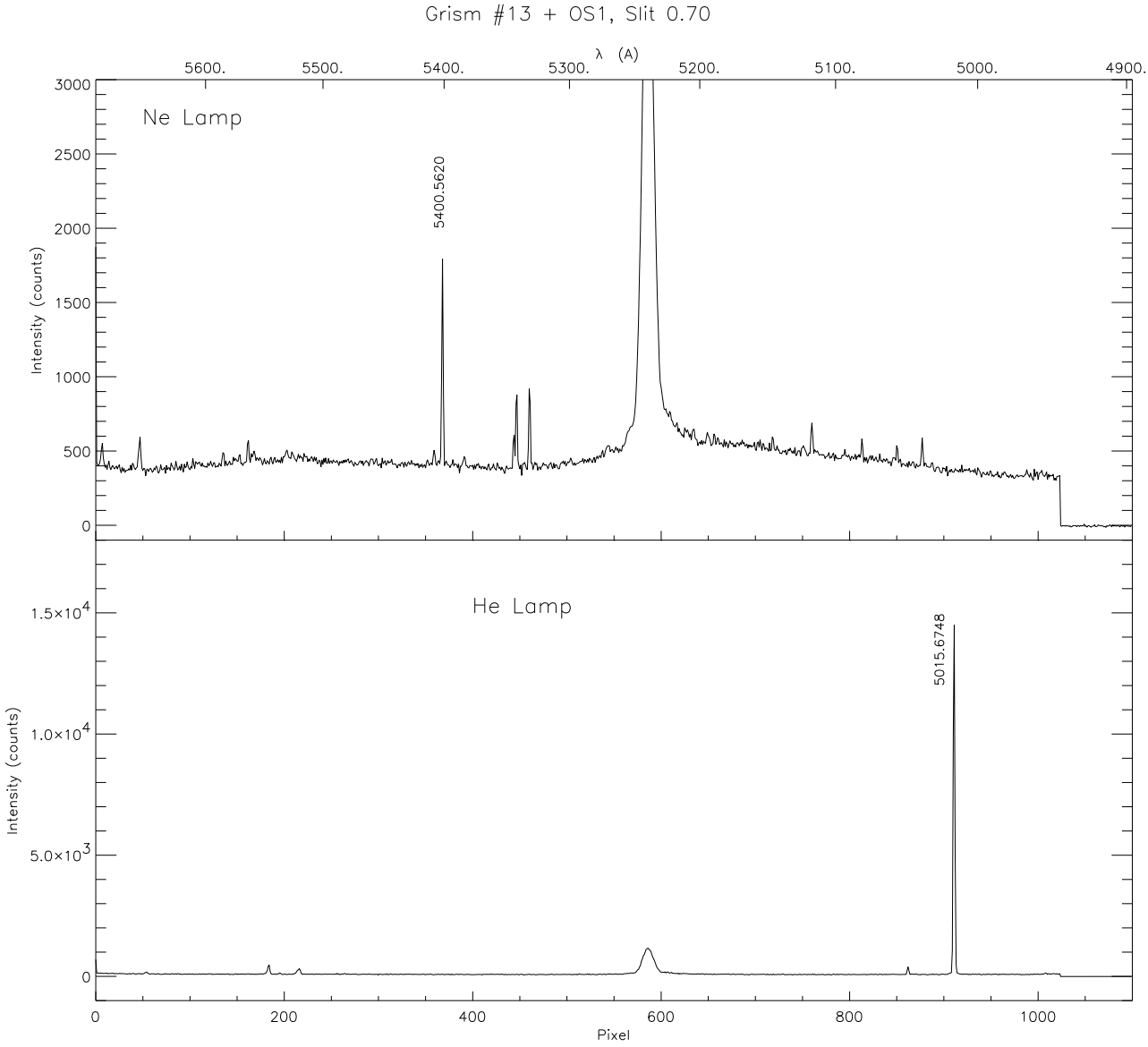


Figure C.14: Calibration spectra of Grism #13 + OS1: Ne and He lamps

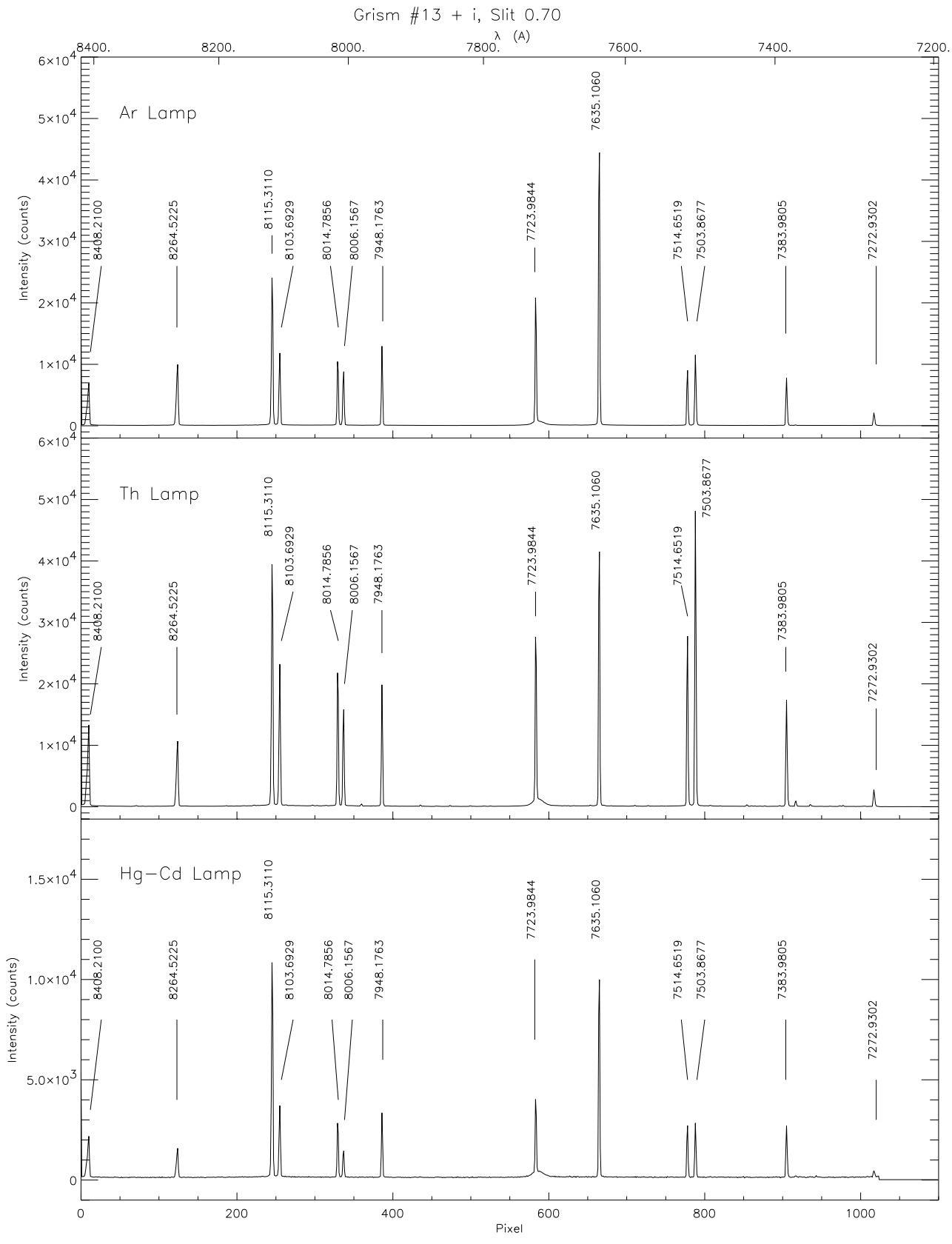


Figure C.15: Calibration spectra of Grism #13 + i: Th, Ar and Hg-Cd lamps

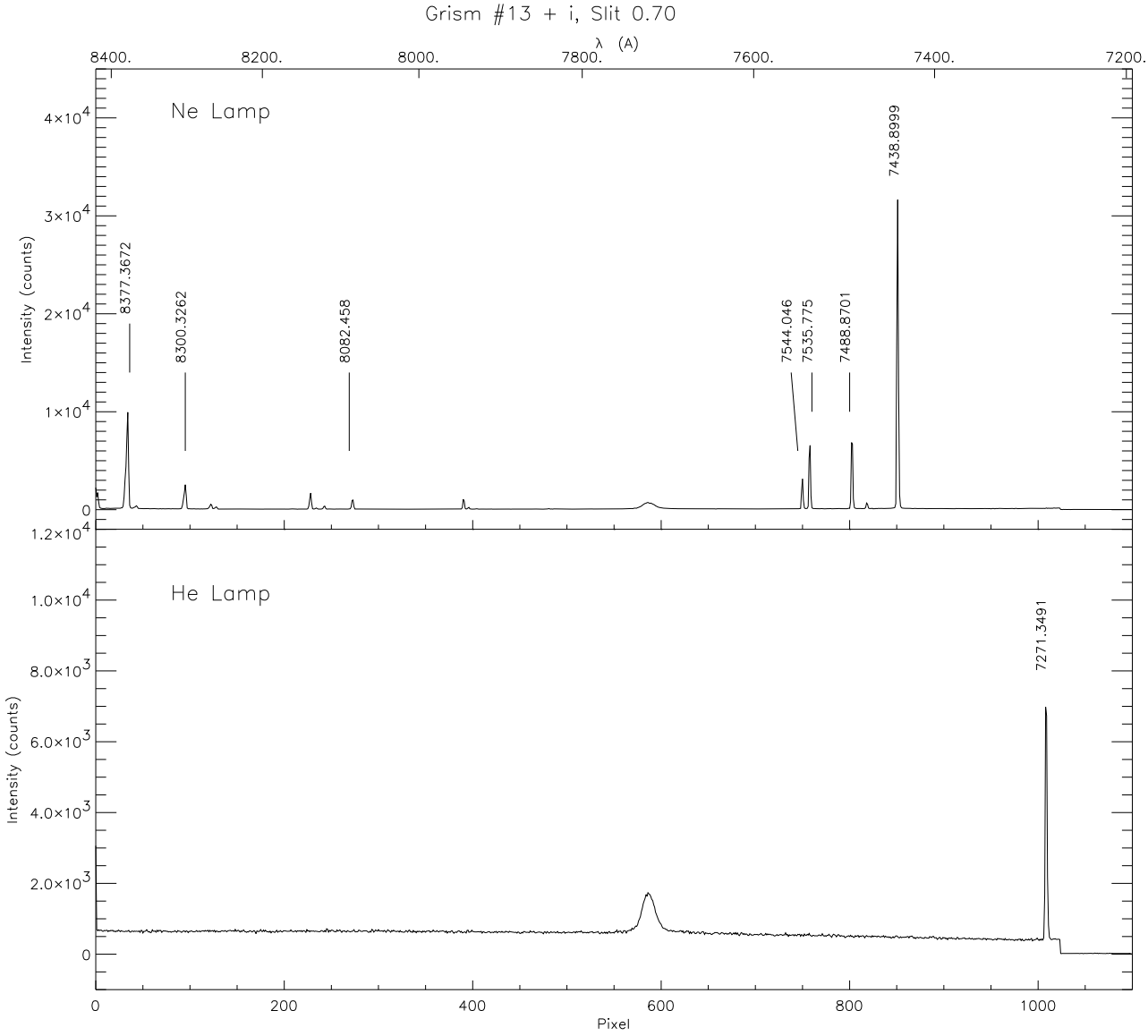


Figure C.16: Calibration spectra of Grism #13 + *i*: Ne and He lamps

C.10 VPH #1

Lamp/Slit	0.70	0.85	1.26	1.69	2.10	3.00	4.22	8.44	Remarks
Ne Ar He Hg-Cd Th	–	–	–	–	–	–	–	–	

Table C.13: Exposure times (in seconds) of the calibration lamps using the VPH1 #1 for the available slits.

Remarks

- Neon Lamp: no lines

Suggested lamps combinations

- Th lamp

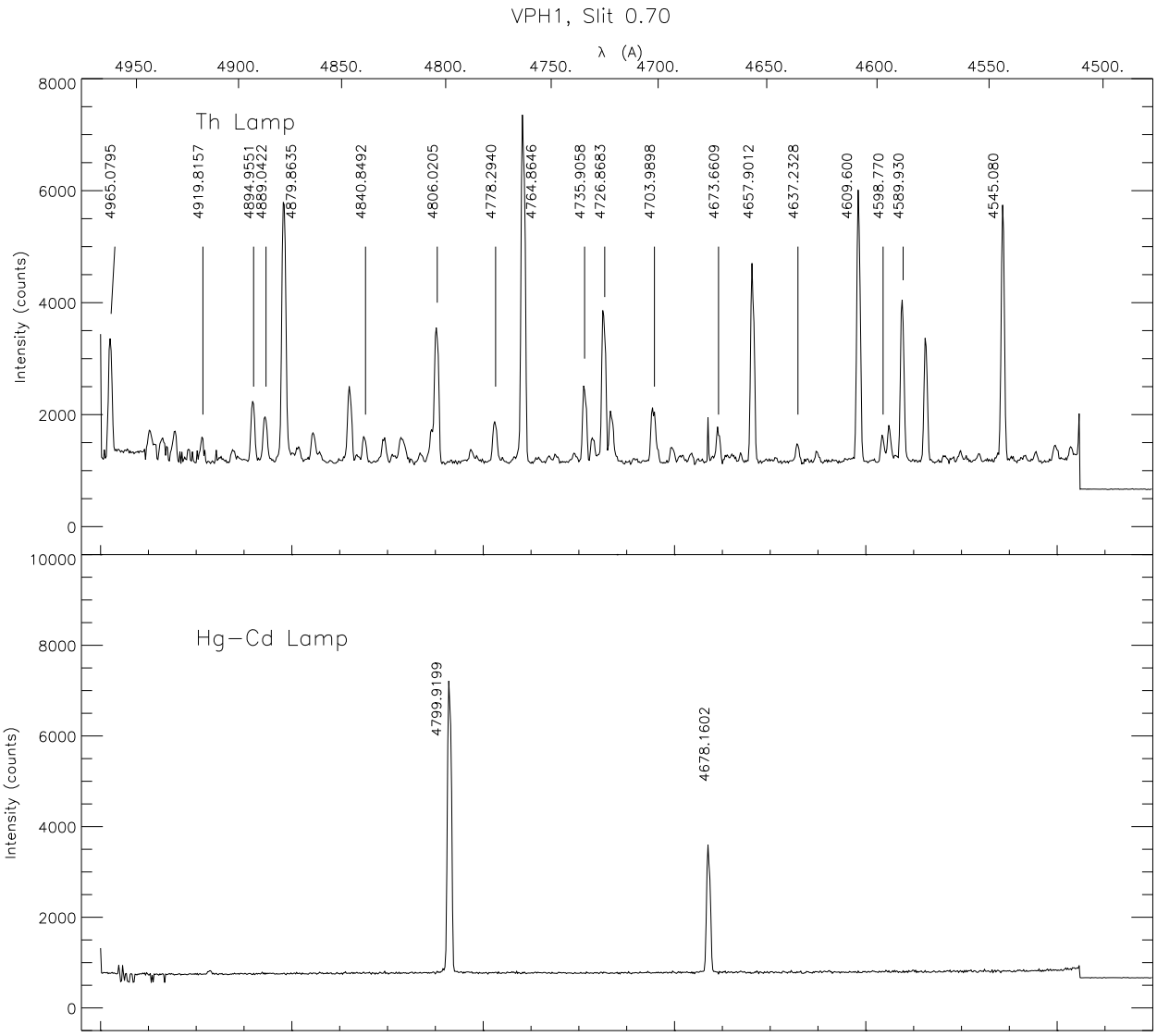


Figure C.17: Calibration spectra of VPH1 #1: Th, and Hg-Cd lamps

C.11 VPH #4

Lamp/Slit	0.70	0.85	1.26	1.69	2.10	3.00	4.22	8.44	Remarks
Ne			40						
Ar									
He									
Hg-Cd			300						
Th			30						

Table C.14: Exposure times (in seconds) of the calibration lamps using the VPH1 #4 for the available slits.

Suggested lamps combinations

- Ne + Th lamps

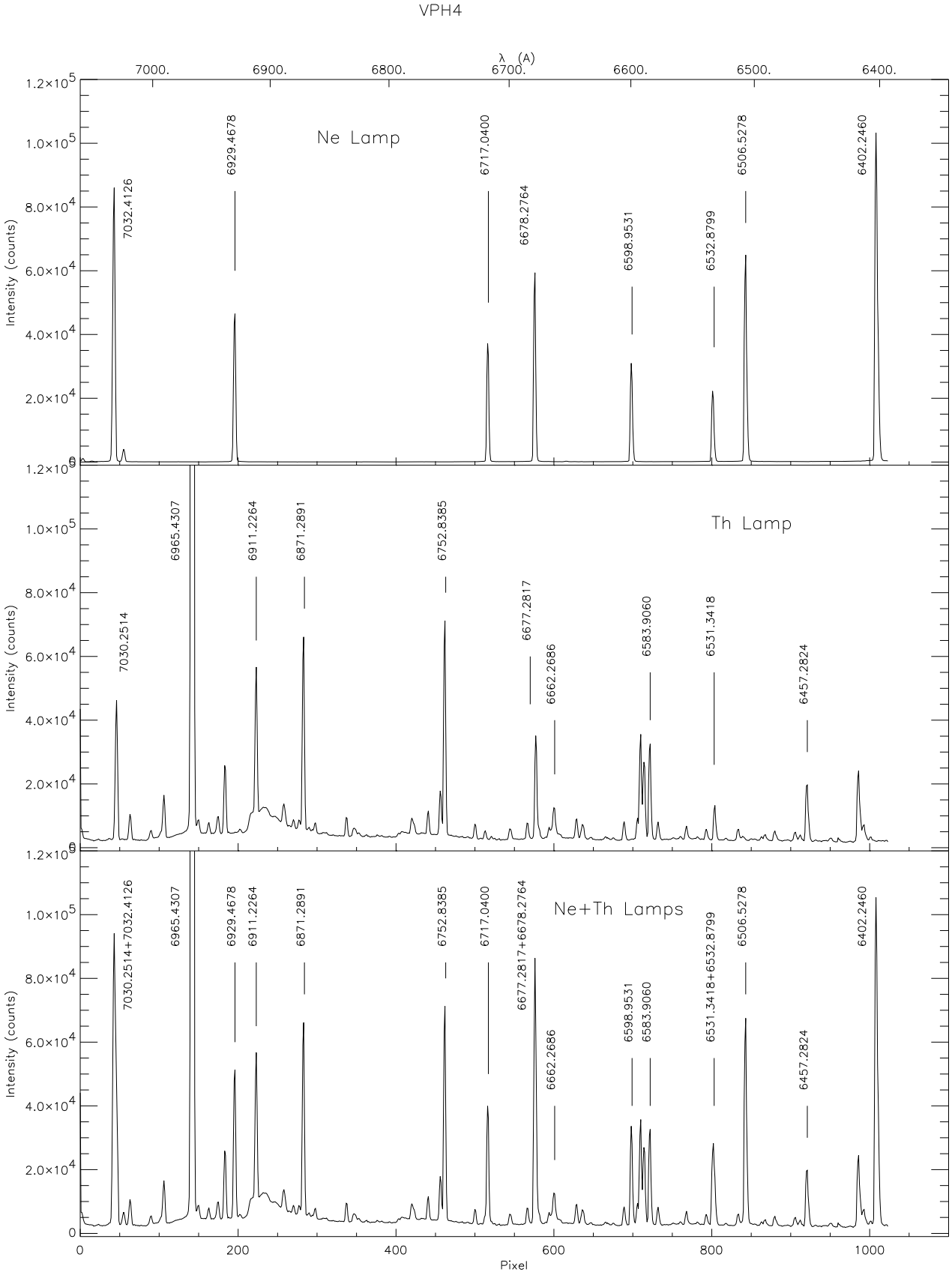


Figure C.18: Calibration spectra of VPH4 #1: Ne, Th, and Ne+Th lamps

C.12 VPH #5

Lamp/Slit	0.70	0.85	1.26	1.69	2.10	3.00	4.22	8.44	Remarks
Ne	180		150		120				
Ar									
He									
Hg-Cd	300		300		300				
Th	20		15		10				

Table C.15: Exposure times (in seconds) of the calibration lamps using the VPH1 #5 for the available slits.

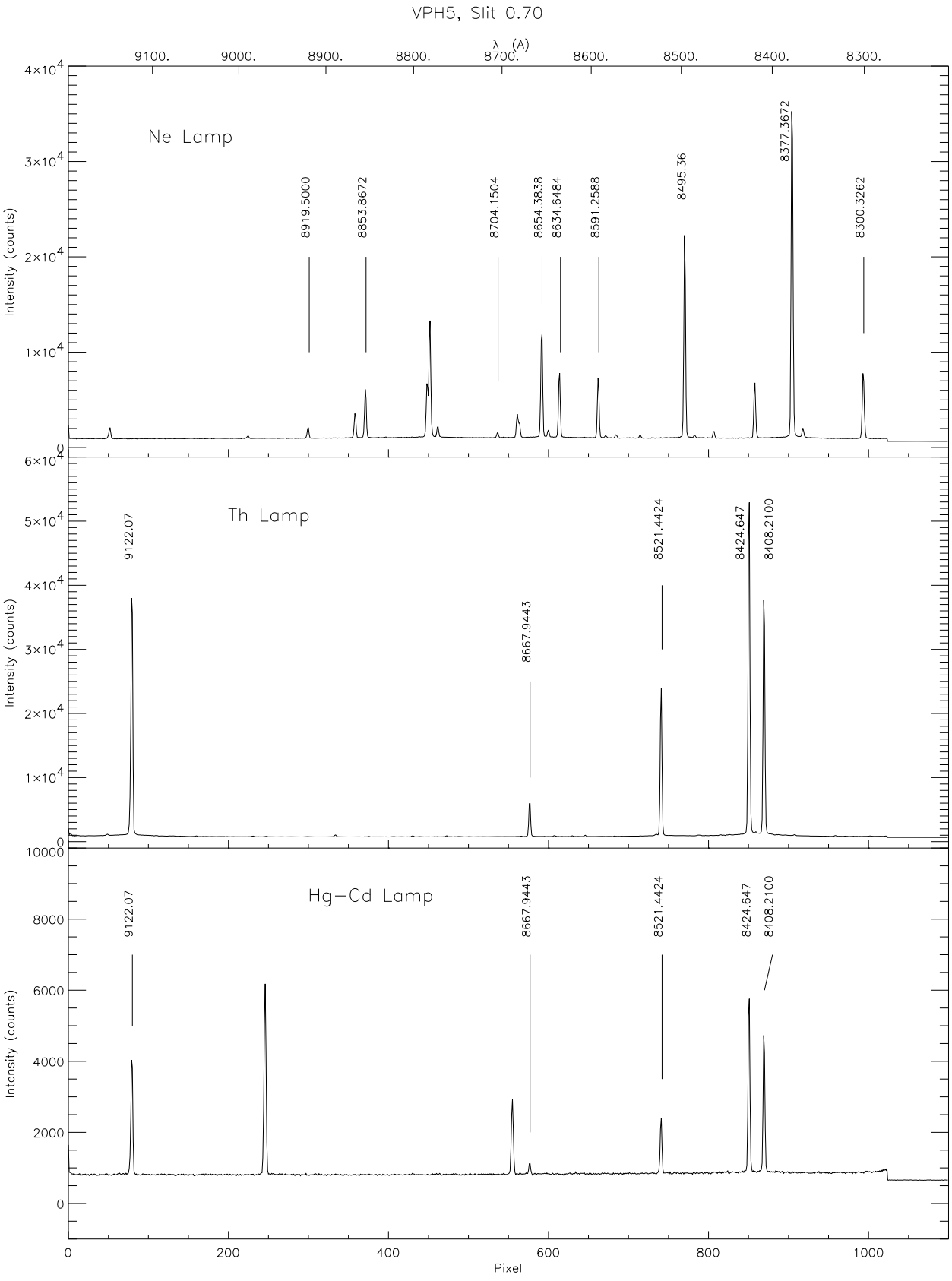


Figure C.19: Calibration spectra of VPH5 #1: Ne, Th and Hg-Cd lamps

C.13 Grism #9 + #10 (Echelle Mode)

Lamp/Slit	Echelle Slit	MF	Remarks
Ne		300	
Ar		300	
He		300	
Hg-Cd		300	
Th		300	

Table C.16: Exposure times (in seconds) of the calibration lamps using the Grism #9 + #10 with the echelle slit.

Remarks

- Atlas of Th lamp includes only relatively unblended lines
- Only Th lamp have a sufficient number of lines for the whole spectral range. Other lamps may be a useful addition in some spectral ranges
- Atlas of 2 very faint orders at blue extreme missing

Suggested lamps combinations

- Th lamp

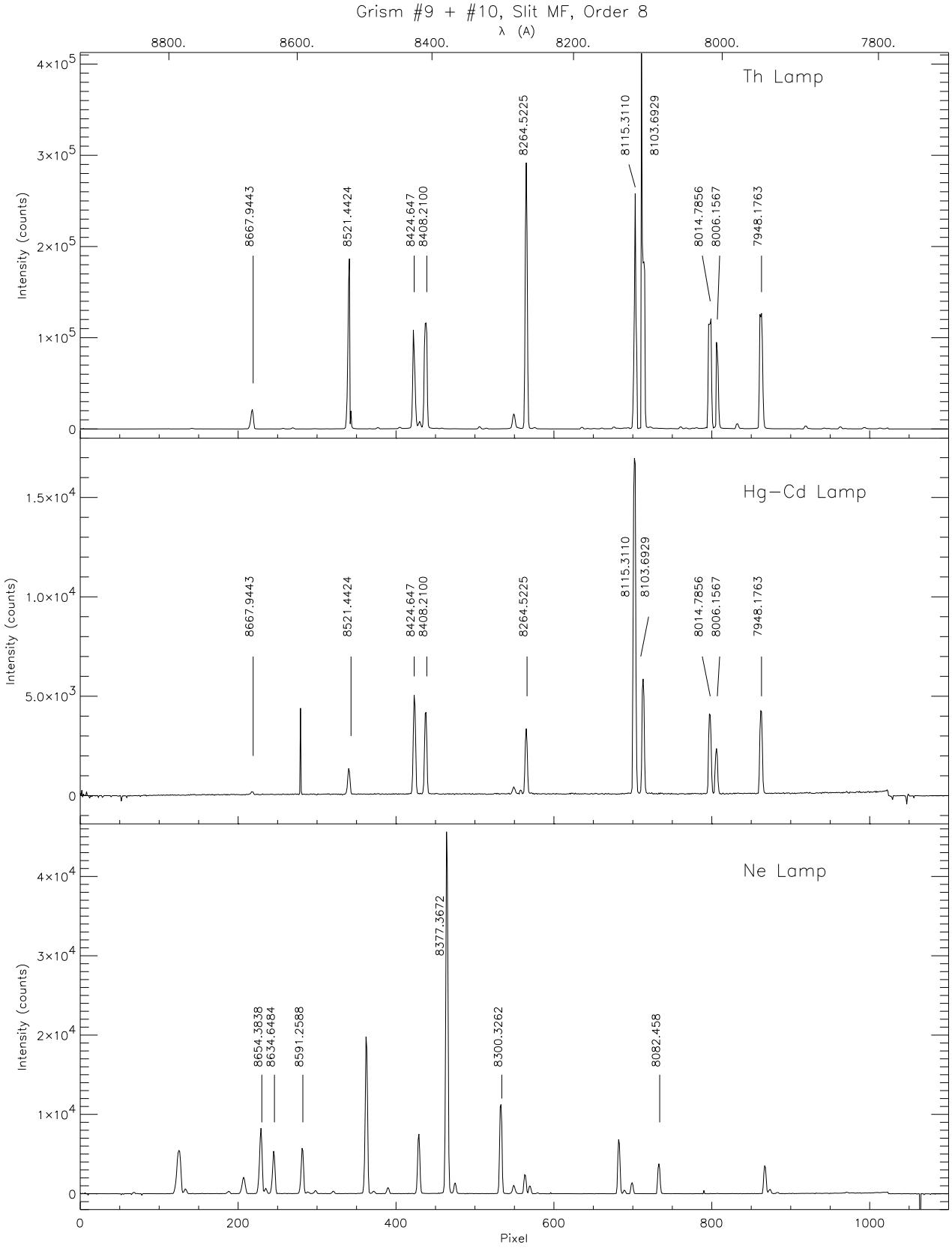


Figure C.20: Calibration spectra of Grism #9 + #10: order 8.

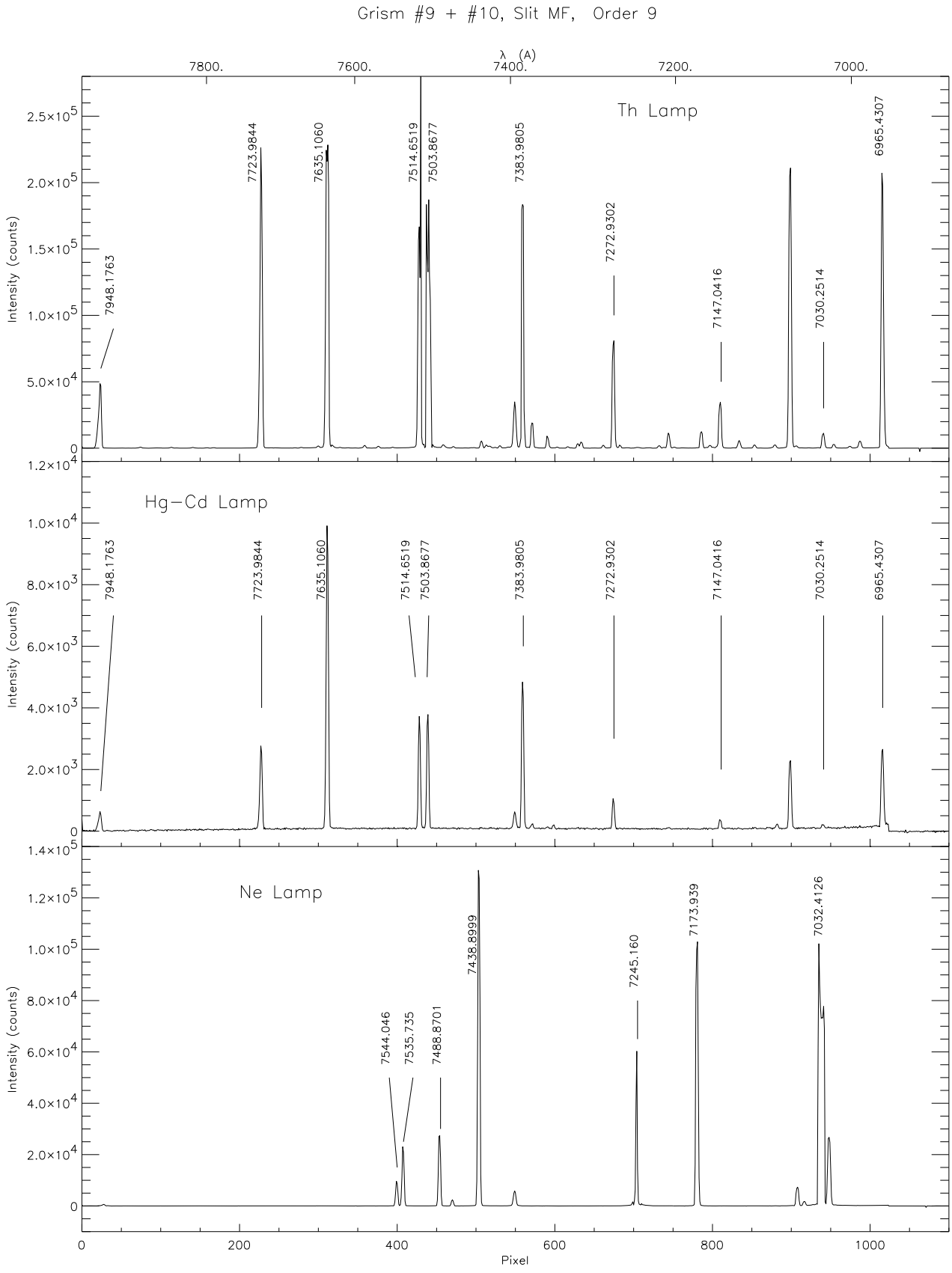


Figure C.21: Calibration spectra of Grism #9 + #10: order 9.

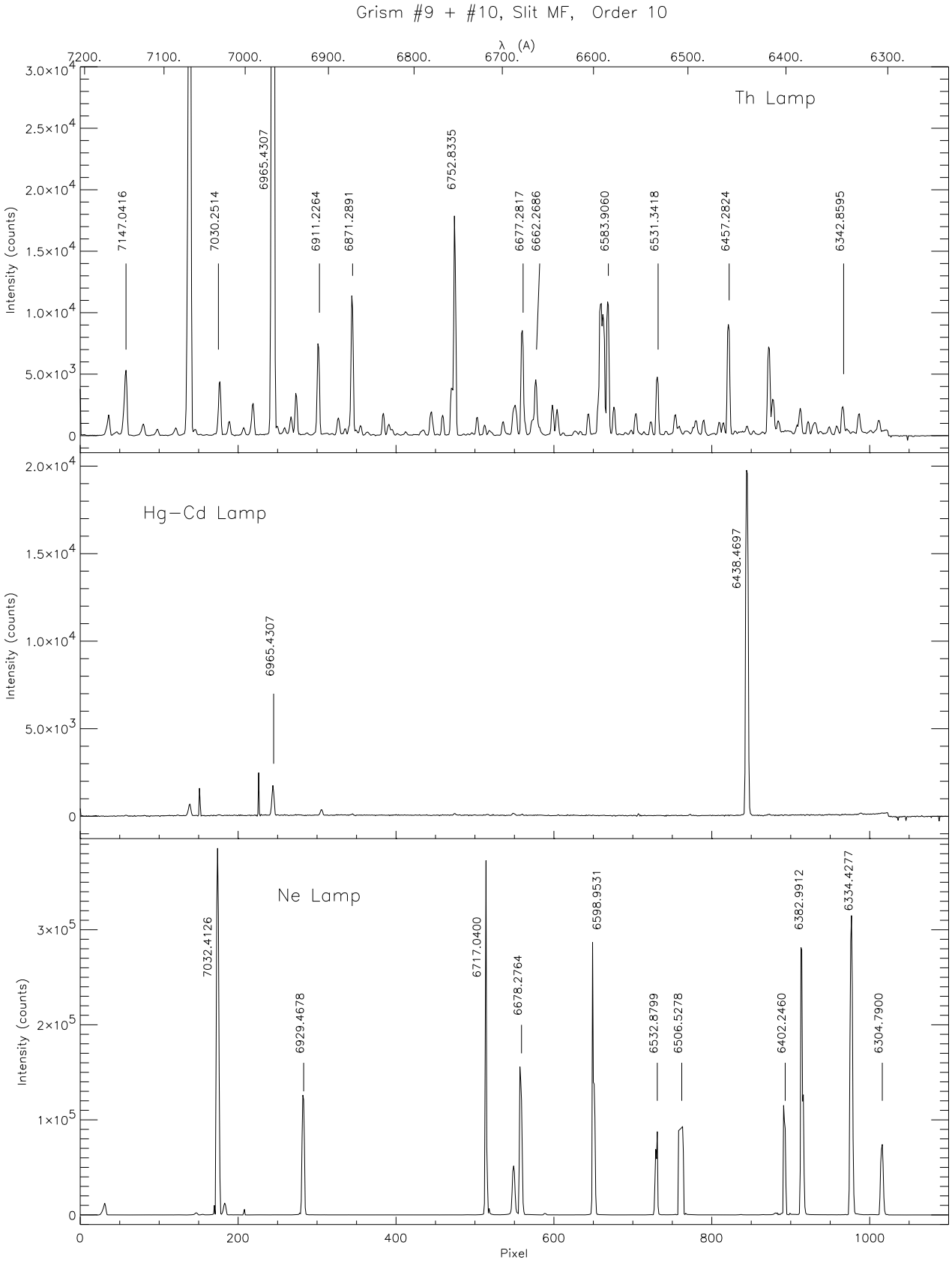


Figure C.22: Calibration spectra of Grism #9 + #10: order 10.

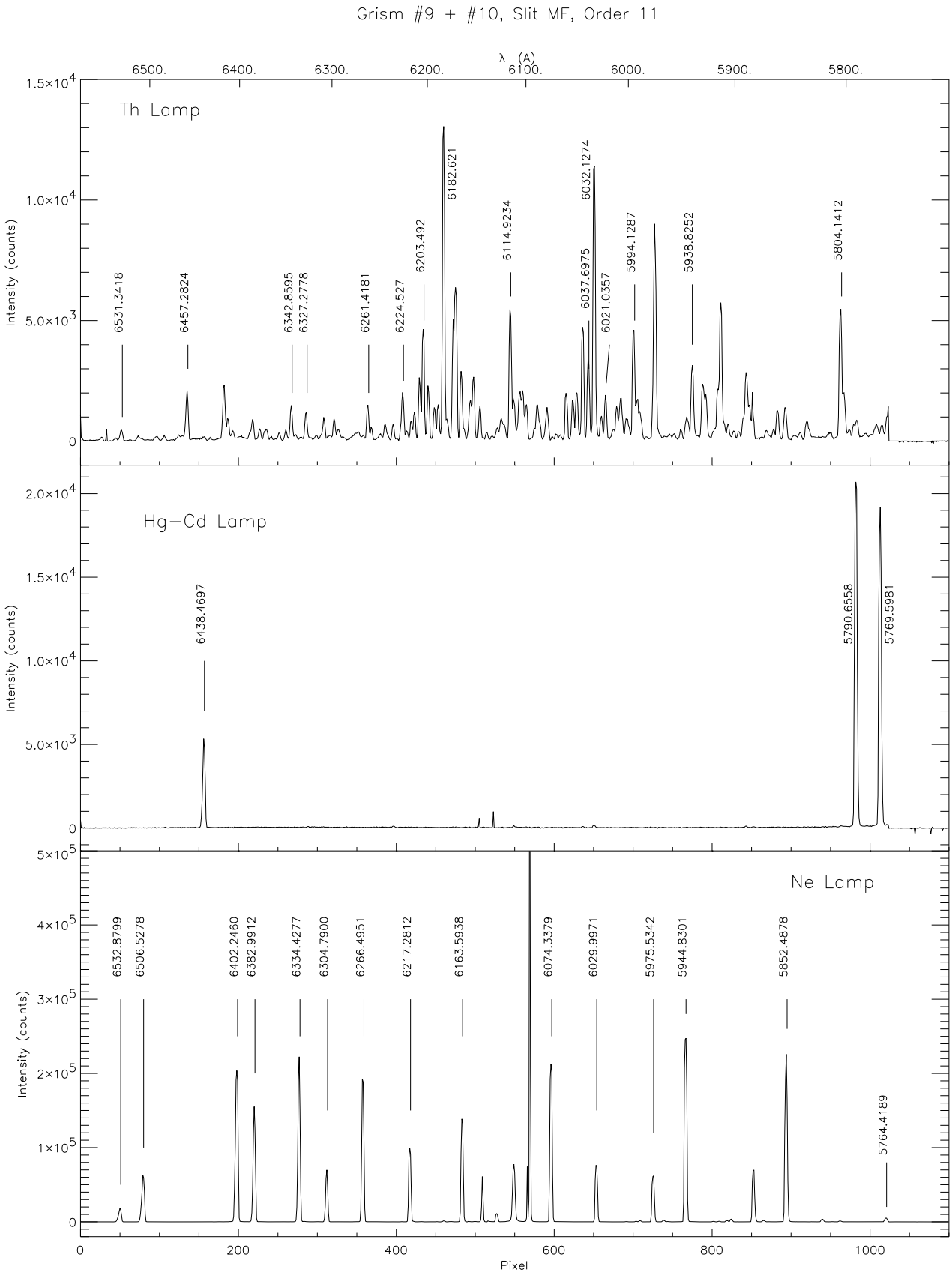


Figure C.23: Calibration spectra of Grism #9 + #10: order 11.

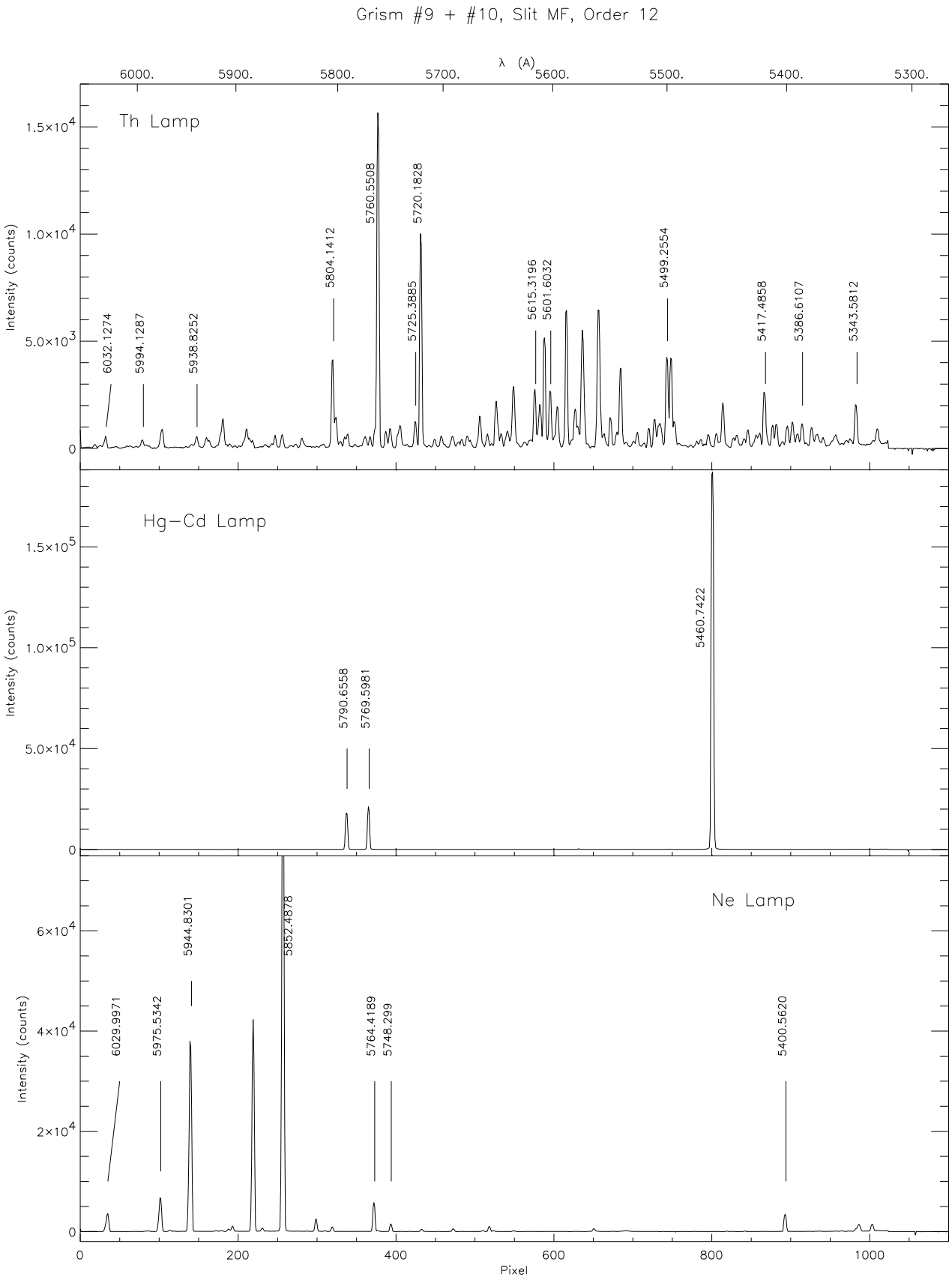


Figure C.24: Calibration spectra of Grism #9 + #10: order 12.

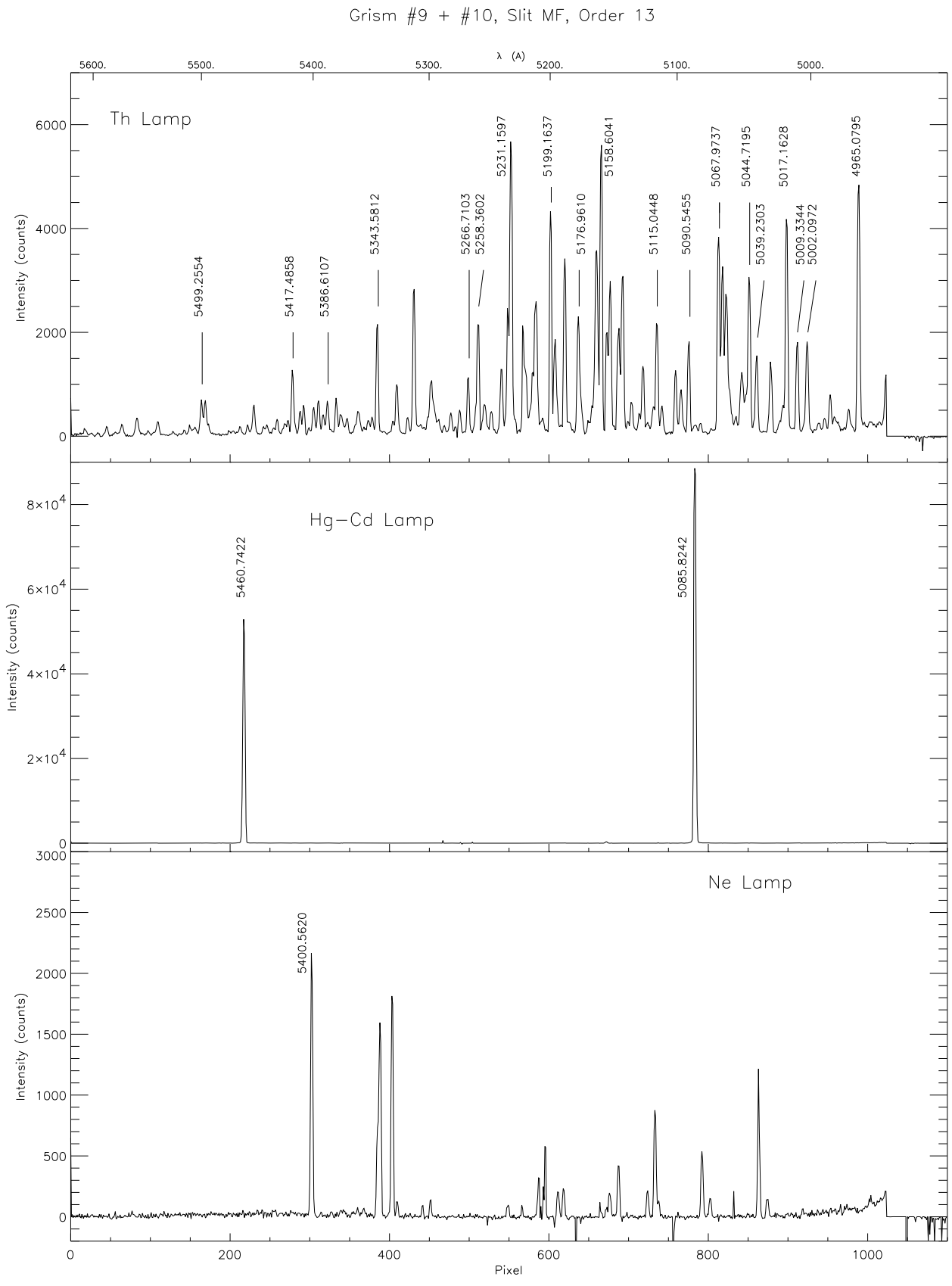


Figure C.25: Calibration spectra of Grism #9 + #10: order 13.

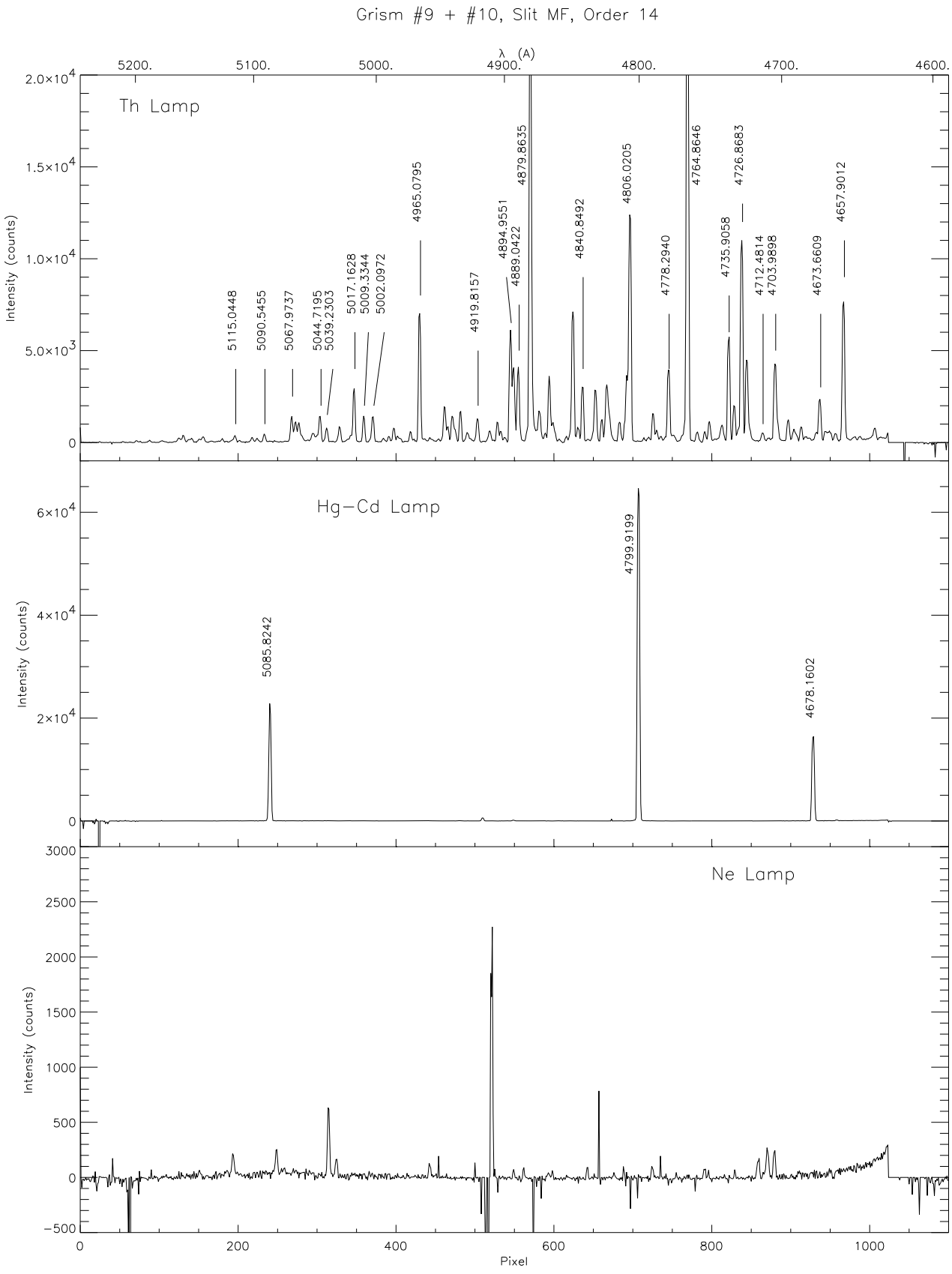


Figure C.26: Calibration spectra of Grism #9 + #10: order 14.

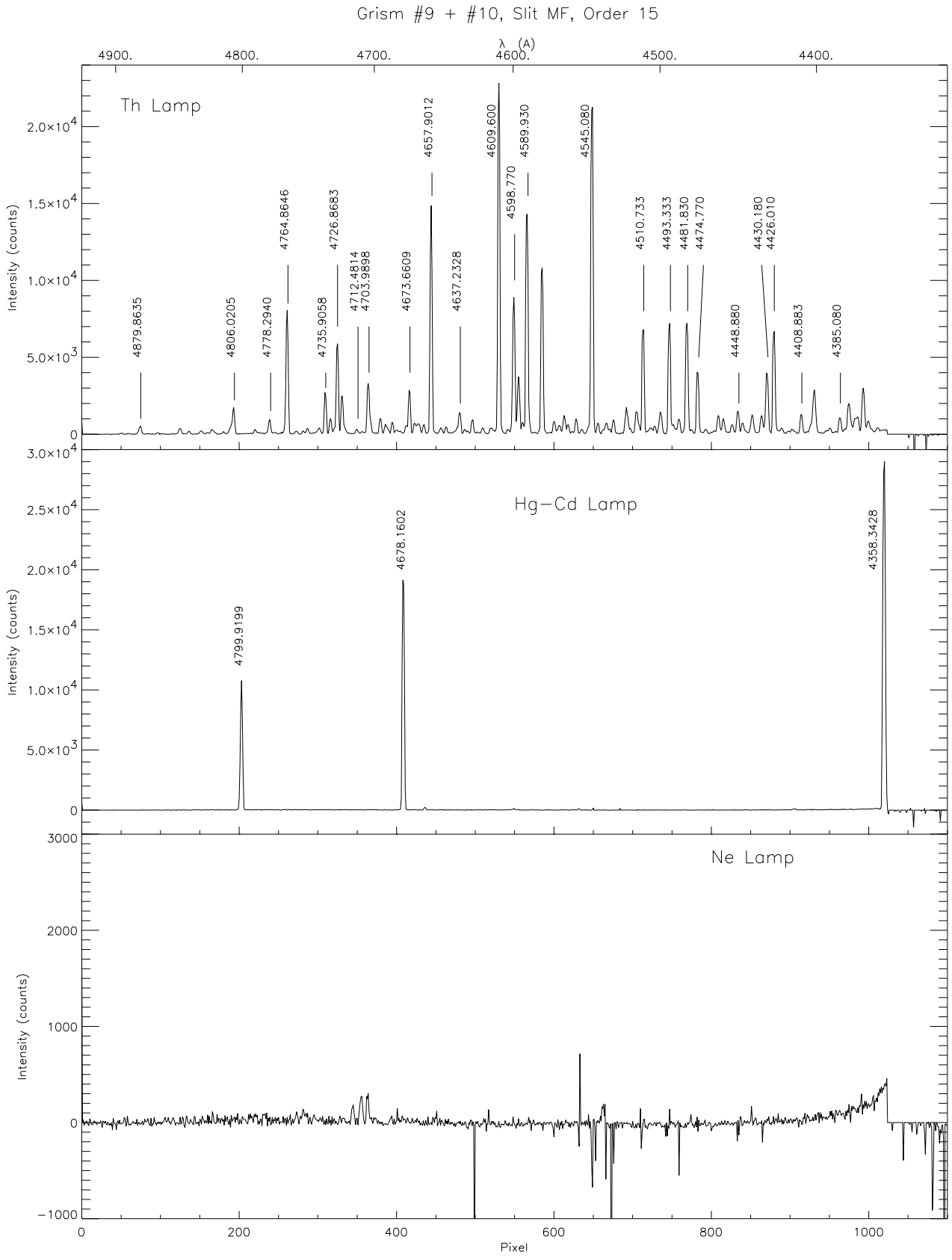


Figure C.27: Calibration spectra of Grism #9 + #10: order 15.

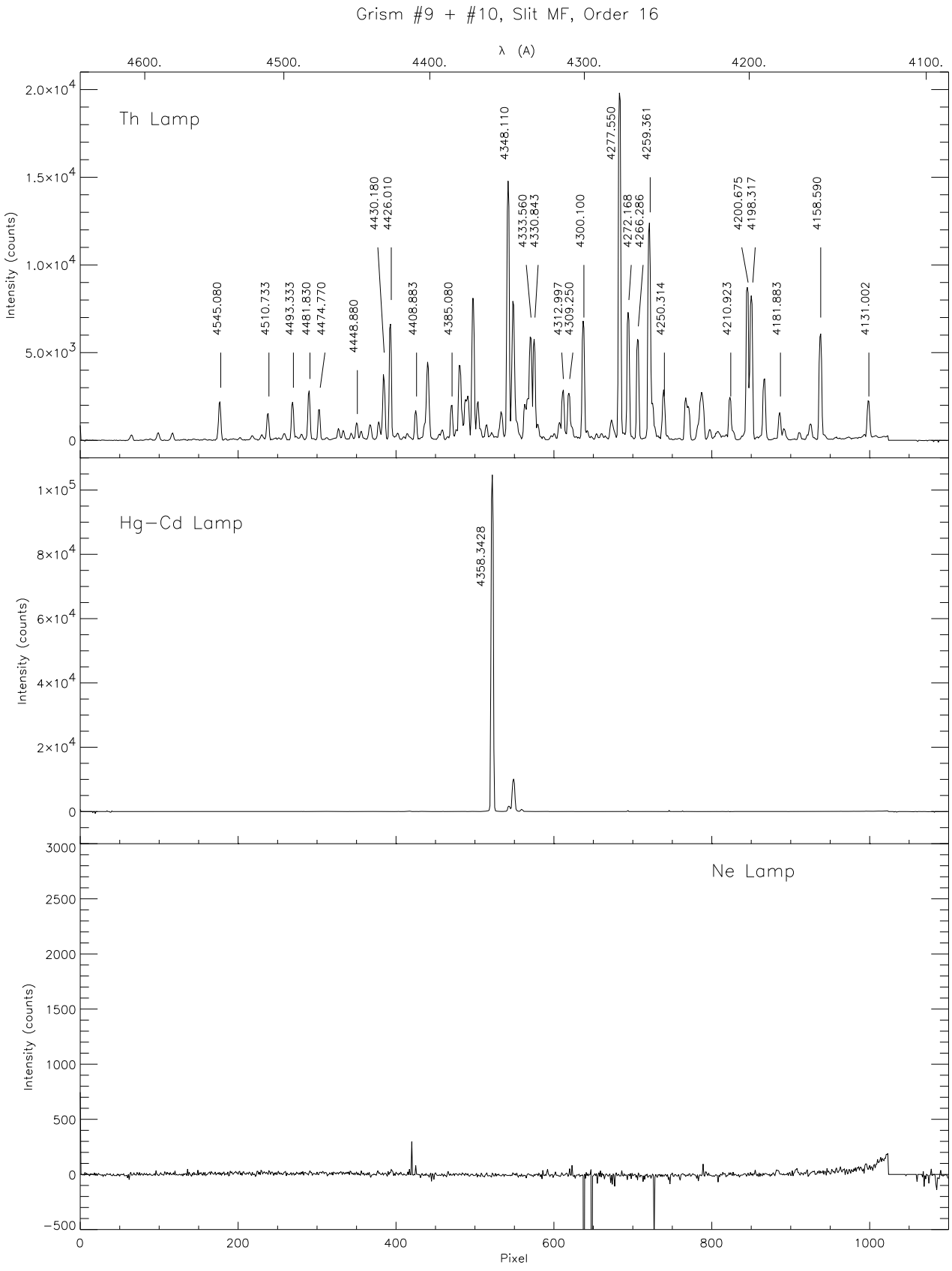


Figure C.28: Calibration spectra of Grism #9 + #10: order 16.

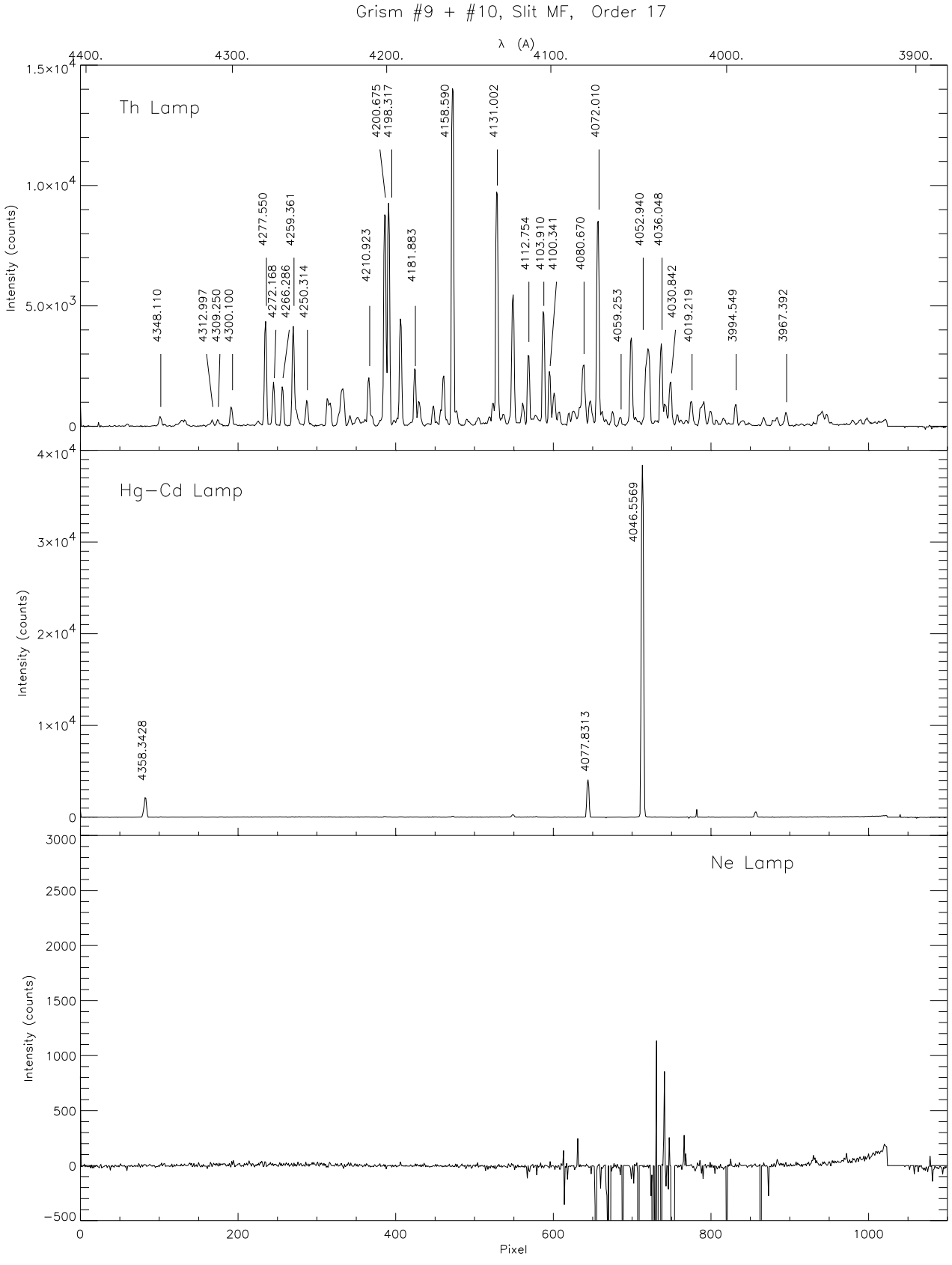


Figure C.29: Calibration spectra of Grism #9 + #10: order 17.

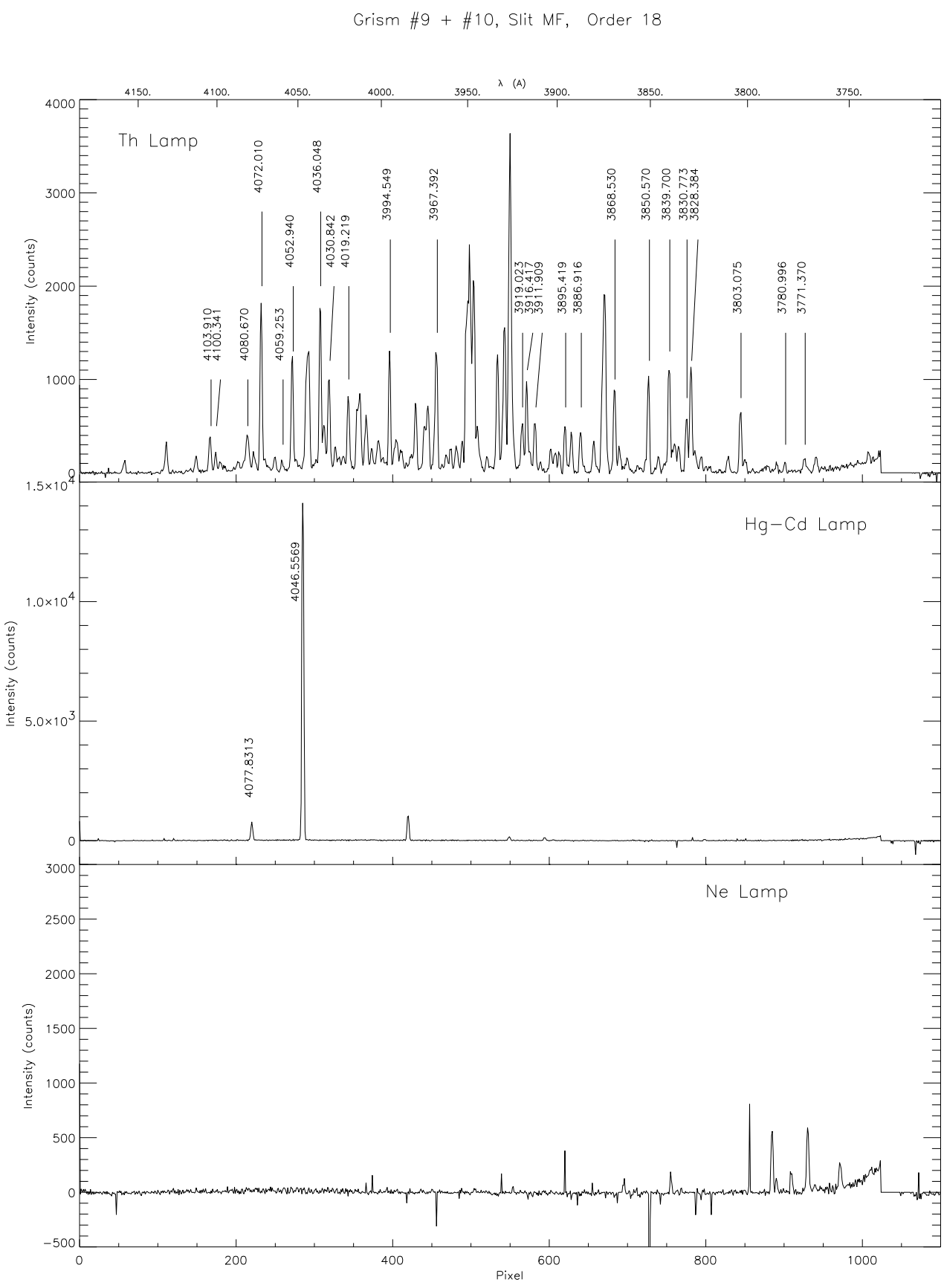


Figure C.30: Calibration spectra of Grism #9 + #10: order 18.

Appendix D

The FITS Header

```
SIMPLE  =                    T / Written by IDL: Tue Jan 8 02:18:51 2002
BITPIX  =                   16 /
NAXIS   =                    2 /
NAXIS1  =                 1100 /
NAXIS2  =                 1100 /
DATE-OBS= '08/01/02'             / Date of data acquisition
DATE    = '2002-01-08'           / Date file was written
EXPSTART= '01:16:46'             / UT at start exposure
JD      = '2452282.55331'        / JD at start exposure
SID-TIME= '09:12:46'             / ST at start exposure
EXPTIME =                 60.0 / exposure time in seconds
RA      = '10:28:01.19'          / Telescope position R.A.
DEC     = '+48:47:56'            / Telescope position Dec
AIRMASS =              2.569295 / Telescope airmass
TELESCOP= 'Mt. Ekar 182 cm. Telescope' /
INSTRUME= 'Afosc Camera'         /
ORIGIN = 'Padova Astronomical Observatory' /
IMAGETYP= 'OBJECT'               /
POSTNRA = '10.466997222222'      / Telescope position R.A.
OBJECT = 'HD90508/sp/pol/g8/0' /
OBSERVER= 'Giro E.'              /
FILTER = 'POL'                   /
GRAT_ANG= ''                     /
GRAT_TYP= 'GR08'                 /
SLIT    = 'SP25'                 /
CAMERA =                  15300 /
POS-ANG =               0.00000 /
DETECTOR= 'tektronix 1kx1k' /
COMMENT1= ''                     /
COMMENT2= ''                     /
COMMENT3= ''                     /
CRPIX1  =                    1 /
CRVAL1  =                    1 /
```

```
CRPIX2  =                    1 /
CRVAL2  =                    1 /
BS01    =          13.022181 /vrd
BS04    =          23.078432 /vdd
BS06    =         0.51316601 /vlg
CCD_TEMP=         -85.569008 /CCD temperature
AFTEMP  = ' 2.4'             /AFOSC temperature (C)
SERRTEMP= ' 0.3'              /Serrourier temperature (C)
M1TEMP  = ' 3.0'             /Primary mirror temperature (C)
OUTTEMP = ' -1.6'             /outdoor temperature (C)
HUMIDITY= ' 25.0'             /Outdoor humidity (%)
END
```

WARNING: The AIRMASS keyword is not reliable

Appendix E

Empty Fields

Field	α (2000)	δ (2000)	Remarks
BLANK 1	04 29 44.9	+54 15 35.5	
9090–7	09 12 00.2	-07 50 47.0	
CT1	10 06 59.3	-02 33 40.0	
CT5	12 28 43.3	-06 55 03.8	
SPKS 01	12 30 38.9	-08 03 27.8	one bright star
CT2	12 57 33.0	-02 23 16.0	
SPKS 11	13 06 55.9	+29 34 47.9	
SER 1	15 15 48.1	-00 42 49.8	
SPKS 12	16 52 33.2	-15 25 56.6	no stars
SPKS 09	18 04 40.0	-04 32 21.4	
SPKS 13	19 21 28.9	+12 27 49.0	
SPKS 14	21 29 34.4	-08 38 30.8	
2345+007	23 48 19.5	+00 57 20.4	
BLANK 6	23 56 40.3	+59 44 59.8	

Table E.1: Empty fields for sky flats.

Appendix F

Target Observability

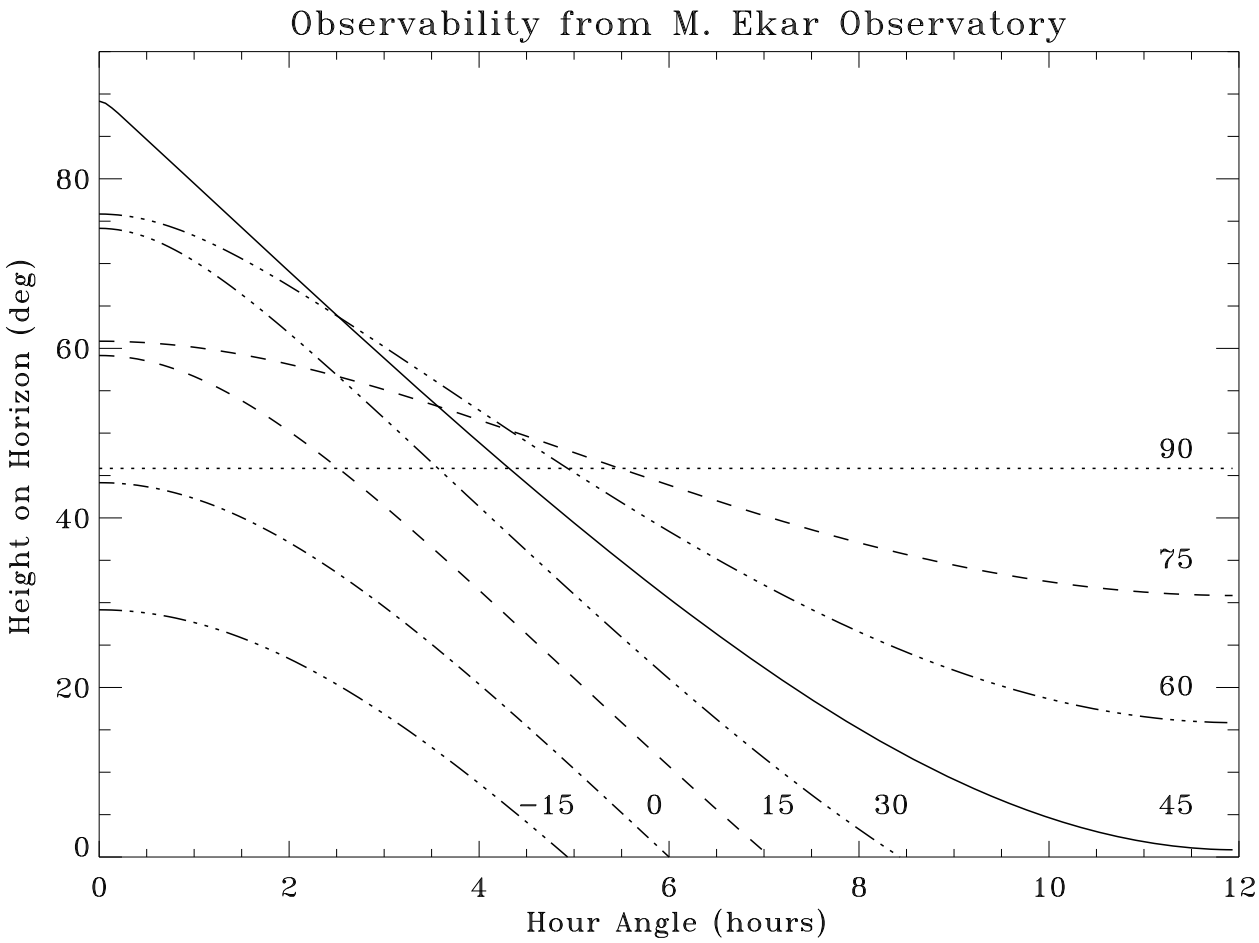


Figure F.1: Height of the horizon at M. Ekar Observatory for targets of declination $\delta = -15, 0, +15, +30, +45, +60, +75, +90$ as a function of the hour angle.

Appendix G

Almanacs and Manuals in Control Room

The following astronomical almanacs, manuals, technical documents are available in control room:

- The AFOSC User Manual (this manual)
- The Astronomical Almanac of the current year
- A list of photometric standards (Landolt 1992)
- A list of spectrophotometric standards
  (from www.eso.org/observing/standards/spectra)
- A list of polarimetric standards
  (from http://www.jach.hawaii.edu/JACpublic/UKIRT/instruments/irpol/irpol_stds.html)
- 9599-3200 Å atlas for the Asiago Echelle spectrograph (Munari & Zwitter 1994)

References

- Andersen M.I., Freyhammer L., Storm J., 1995, in *Proceedings of an ESO/ST-ECF workshop on calibrating and understanding HST and ESO instruments* P. Benvenuti ed., pag. 87
- Barbieri C. & Galazzi L., 1973, in *Atti delle celebrazioni del V Centenario della nascita di Nicoló Copernico*
- Bonanno G., Bruno P., Cosentino R., Scuderi S., Bortoletto F., D'Alessandro M., Bonoli C., Fantinel D., Carbone A., Evola, G., 2000, in *Optical Detectors for Astronomy II: State-of-the-Art at the Turn of the Millenium. 4th ESO CCD Workshop*, Amico P. and Beletic J.W. eds., pag. 389
- Claudi R.U., 1998, Document AFOSC-007
- Giro E., Pernechele C., Desidera S., Chiomento V., Frigo A., Traverso L., Molinari E., Conconi P., Crimi G., Bianco A., 2002, *First tests with VPH Grisms from AFOSC*, Padova and Asiago Observatories, Technical Report n. 20
- Landolt A.U., 1992, AJ 104, 340
- Munari U., 1988, Technical Report, Padova and Asiago Observatories
- Munari U. & Zwitter, 1994, *9599-3200 Å atlas for the Asiago Echelle spectrograph*, Technical Report n.4, Padova and Asiago Observatories
- Pernechele C., Bortoletto F., Fantinel D., Giro E., 2000, PASP 112, 996
- Pernechele C. & Giro E., 2002, in Astronomical Telescopes and Instrumentation, SPIE Proc., in press

Acknowledgements

We thank Claudio Pernechele for the supervision of this work and the contributions on AFOSC optics; Maurizio D'Alessandro for many useful suggestions for CCD testing; Riccardo Claudi, who provided "historical" documentation on AFOSC; S. Benetti, R. Falomo and D. Bettoni for a careful reading of an earlier version of the manual and for many suggestions; S. Benetti for providing the list of empty fields; Alessandro Pizzella for providing the original files of DFOSC manual.

The whole M. Ekar Observatory staff (V. Chiomento, L. Contri, A. Frigo, A. Gianesini, G. Lessio, G. Martorana., M. Rebeschini, I. Rigoni, L. Rigoni, I. Stefani, D. Strazzabosco, L. Traverso) gave continuous support during the tests described in this manual.

Many observers gave useful suggestions about the content of this manual during its preparation.